Wind Energy

- https://greencoast.org/onshore-vs-offshore-wind/
- https://www.brunel.net/en/blog/renewable-energy/onshore-offshore-wind
- https://en.wikipedia.org/wiki/Wind_farm

Origin of Wind

- http://www.ces.fau.edu/nasa/content/resources/global-wind-patterns.php#:~:text=Large%20global%20wind%20systems%20are,drive%20the%20oceans'%20surface%20currents.&text=Warm%20air%20rises%20at%20the,it%20is%20not%20this%20simple.
- https://www.ck12.org/earth-science/local-winds/lesson/Local-Winds-MS-ES/#:~:text=Local%20winds%20are%20winds%20that,range%20can%20affect%20local%20winds.
- https://byjus.com/physics/types-of-wind/

Wind turbine control

- http://xn--drmstrre-64ad.dk/wp-content/wind/miller/windpower%20web/en/tour/wtrb/size.htm
- https://www.ni.com/en-in/innovations/white-papers/08/wind-turbine-control-methods.html#section--1572769063

Advantages of Wind Energy

1) Wind Energy is an inexhaustible source of energy and is virtually a limitless resource.

2) Energy is generated without polluting environment.

3) This source of energy has tremendous potential to generate energy on large scale.

4) Like solar energy and hydropower, wind power taps a natural physical resource.

5) Windmill generators **don't** emit any emissions that can lead to acid rain or greenhouse effect.

6) Wind Energy can be used directly as mechanical energy.

7) In remote areas, wind turbines can be used as great resource to generate energy.

8) In combination with Solar Energy they can be used to provide reliable as well as steady supply of electricity.

9) Land around wind turbines can be used for other uses, e.g. Farming.

Disadvantages of Wind Energy

1) Wind energy requires expensive storage during peak production time.

2) It is unreliable energy source as winds are uncertain and unpredictable.

3) There is visual and aesthetic impact on region.

4) Requires large open areas for setting up wind farms.

5) Noise pollution problem is usually associated with wind mills.

6) Wind energy can be harnessed only in those areas where wind is strong enough and weather is windy for most parts of the year.

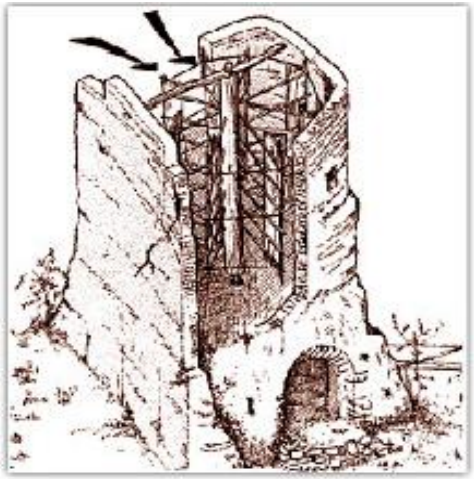
7) Usually places, where wind power set-up is situated, are away from the places where demand of electricity is there. Transmission from such places increases cost of electricity.

8) The average efficiency of wind turbine is very less as compared to fossil fuel power plants. We might require many wind turbines to produce similar impact.

9) It can be a threat to wildlife. Birds do get killed or injured when they fly into turbines.

10) Maintenance cost of wind turbines is high as they have mechanical parts which undergo wear and tear over the time.

Brief History – Early Sys



Harvesting wind power isn't exactly a new idea – sailing ships, wind-mills, wind-pumps

1st Wind Energy Systems

- Ancient Civilization in the Near East / Persia
- Vertical-Axis Wind-Mill: sails connected to a vertical shaft connected to a grinding stone for milling

Wind in the Middle Ages

- Post Mill Introduced in Northern Europe
- Horizontal-Axis Wind-Mill: sails connected to a horizontal shaft on a tower encasing gears and axles for translating horizontal into rotational motion

Wind in 19th century US

- Wind-rose horizontal-axis water-pumping wind-mills found throughout rural America

Torrey, Volta (1976) Wind-Catchers: American Windmills of Yesterday and Tomorrow. Stephen Green Press, Vermont.
Righter, Robert (1996) Wind Energy in America. University of Oklahoma Press, Oklahoma.

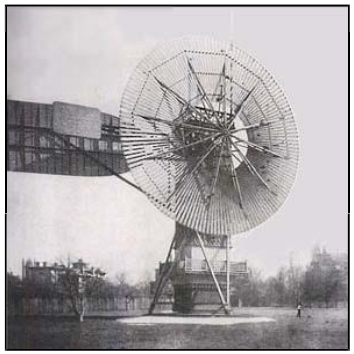
Brief History - Rise of Wind Powered Electricity

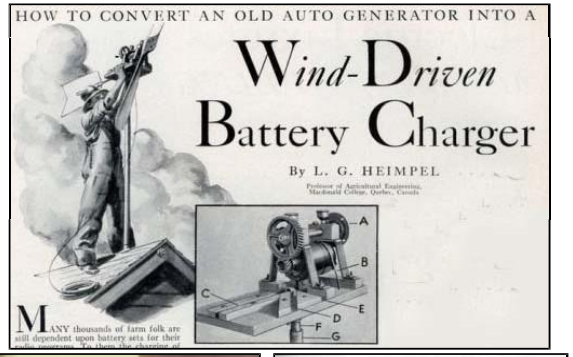
1888: Charles Brush builds first large-size wind electricity generation turbine (17 m diameter wind rose configuration, 12 kW generator)

1890s: Lewis Electric Company of New York sells generators to retro-fit onto existing wind mills

1920s-1950s: Propeller-type 2 & 3-blade horizontal-axis **wind electricity conversion systems** (WECS)

1940s – 1960s: Rural Electrification in US and Europe leads to decline in WECS use




HOW TO CONVERT AN OLD AUTO GENERATOR INTO A

Wind-Driven Battery Charger

By L. G. HEIMPEL

Torrey, Volta (1976) Wind-Catchers: American Windmills of Yesterday and Tomorrow. Stephen Green Press, Vermont.
Righter, Robert (1996) Wind Energy in America. University of Oklahoma Press, Oklahoma.

Brief History

Key attributes of this period:

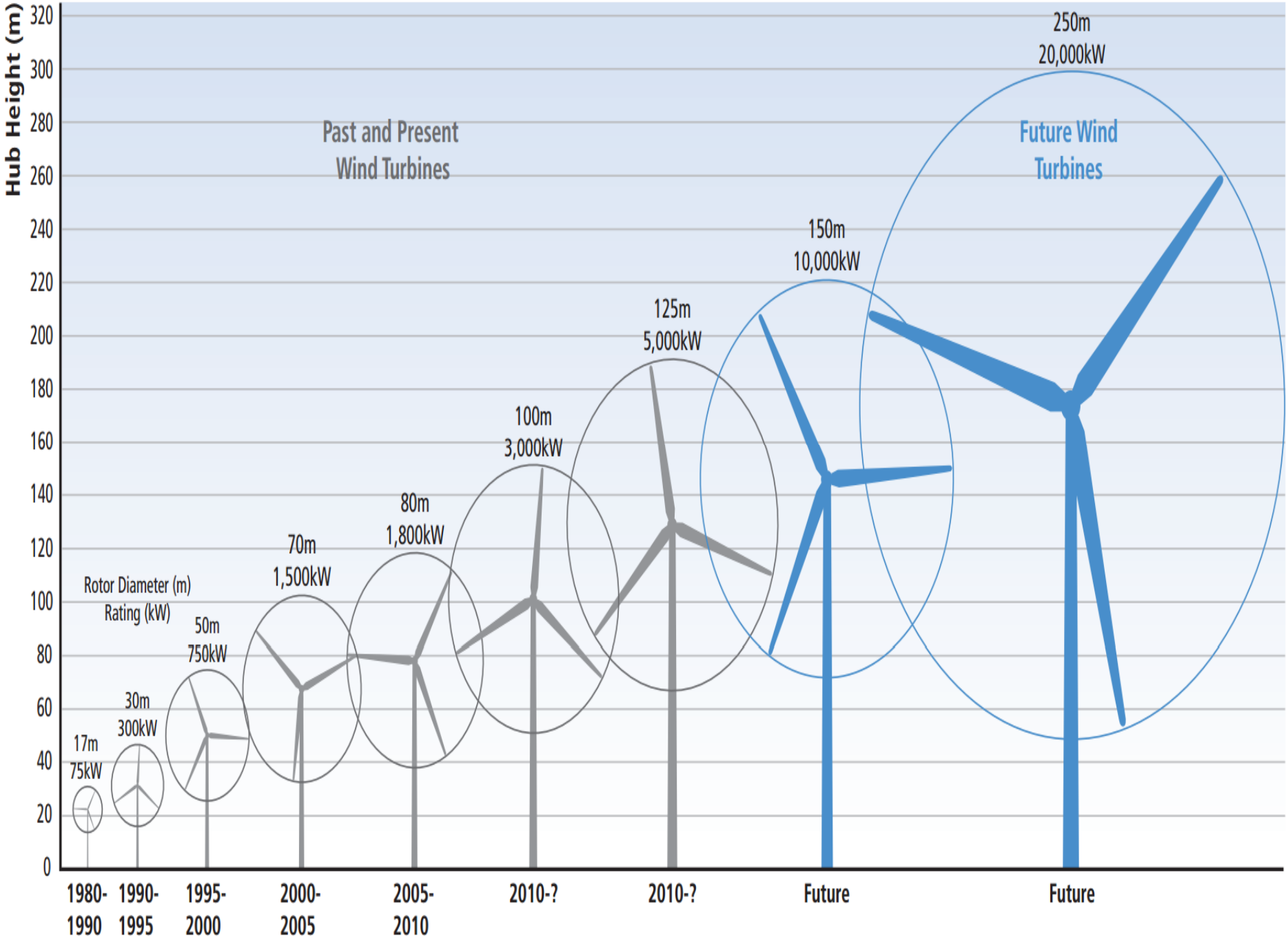
- Scale increase
- Commercialization
- Competitiveness
- Grid integration

Catalyst for progress: OPEC Crisis (1970s)

- Economics
- Energy independence
- Environmental benefits

Turbine Standardization:

- 3-blade Upwind Horizontal-Axis
- on a monopole tower


Hub Height (m)
320
300
280
260
240
220
200
180
160
140
120
100
80
60
40
20
0
Past and Present
Wind Turbines
Future Wind
Turbines
Rotor Diameter (m)
Rating (kW)
17m
75kW
30m
300kW
50m
750kW
70m
1,500kW
80m
1,800kW
100m
3,000kW
125m
5,000kW
150m
10,000kW
250m
20,000kW
1980-
1990
1990-
1995
1995-
2000
2000-
2005
2005-
2010
2010-?
2010-?
Future
Future

Origin of Wind

Wind – Atmospheric air in motion

Energy source

Solar radiation differentially absorbed by earth surface converted through convective processes due to temperature differences to air motion

Spatial Scales

Planetary scale: global circulation

Synoptic scale: weather systems

Meso scale: local topographic or thermally induced circulations

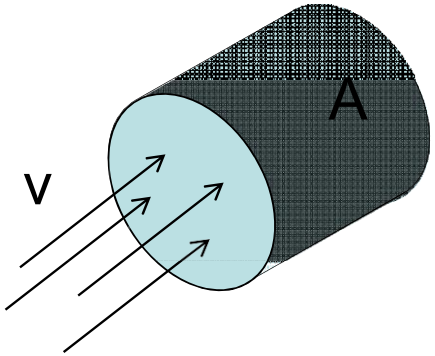
Micro scale: urban topography

Source for Graphic: NASA / GSFC

Fundamental Equation of Wind Power

- Wind Power depends on:
  - amount of air (volume)
  - speed of air (velocity)
  - mass of air (density)

  flowing through the area of interest (flux)



- **Kinetic Energy** definition:
  - $KE = \frac{1}{2} * m * v^2$
- Power is KE per unit time:
  - $P = \frac{1}{2} * \dot{m} * v^2$

$$\dot{m} = \frac{dm}{dt}$$ mass flux

- Fluid mechanics gives **mass flow rate** (density * volume flux):
- $dm/dt = \boldsymbol{\rho} * A * v$
- Thus:
  - $P = \frac{1}{2} * \boldsymbol{\rho} * A * v^3$

- Power ~ cube of velocity
- Power ~ air density
- Power ~ rotor swept area $A = \pi r^2$

Wind Speed Measuring Instrumentation

- Momentum transfer – cups, propellers, and pressure plates
- Pressure on stationary sensors – pilot tubes and drag spheres
- Heat transfer – hot wires and hot films
- Doppler effects – acoustics and laser
- Special methods – ion displacement, vortex shedding, etc

Cup Anemometers

- Cup anemometers use their rotation, which varies in proportion to the wind speed, to generate a signal.
- The photoelectric switch type has a disc containing up to 120 slots and a photocell.
- The periodic passage of the slots produces pulses during each revolution of the cup.
- The response and accuracy of a cup anemometer are determined by its weight, physical dimensions, and internal friction.
- By changing any of these parameters, the response of the instrument will vary.
- If turbulence measurements are desired, small, light weight, low-friction sensors should be used.
- Typically, the most responsive cups have a distance constant of about1 m.
- Where turbulence data are not required, the cups can be larger and heavier, with distance constants from 2 to 5 m.
- Typical accuracy values (based on wind tunnel tests) for cup anemometers are about 2%.

- Environmental factors can affect cup anemometers and reduce their reliability.
- These include ice or blowing dust.
- Dust can lodge in the bearings, causing an increase in friction and wear and reducing anemometer wind speed readings.
- If an anemometer ices up, its rotation will slow, or completely stop, causing erroneous wind speed signals, until the sensor thaws completely.
- Heated cup anemometers can be used, but they require a significant source of power.
- Because of these problems, the assurance of reliability for cup anemometers depends on calibration and service visits.
- The frequency of these visits depends on the site environment and the value of the data.

Propeller Anemometers

- Propeller anemometers use the wind blowing into a propeller to turn a shaft that drives an AC or DC (most common) generator, or a light chopper to produce a pulse signal.
- The designs used for wind energy applications have a fast response and behave linearly in changing wind speeds.
- In a typical horizontal configuration, the propeller is kept facing the wind by a tail-vane, which also can be used as a direction indicator. The accuracy of this design is about 2%, similar to the cup anemometer.
- The propeller is usually made of polystyrene foam or polypropylene. The problems of reliability of propeller anemometers are similar to those discussed for cup anemometers.
- When mounted on a fixed vertical arm, the propeller anemometer may be used for measuring the vertical wind component.
- A configuration for measuring three components of wind velocity
- The propeller anemometer responds primarily to wind parallel to its axis, and the wind perpendicular to the axis has no effect.

Sonic Anemometers

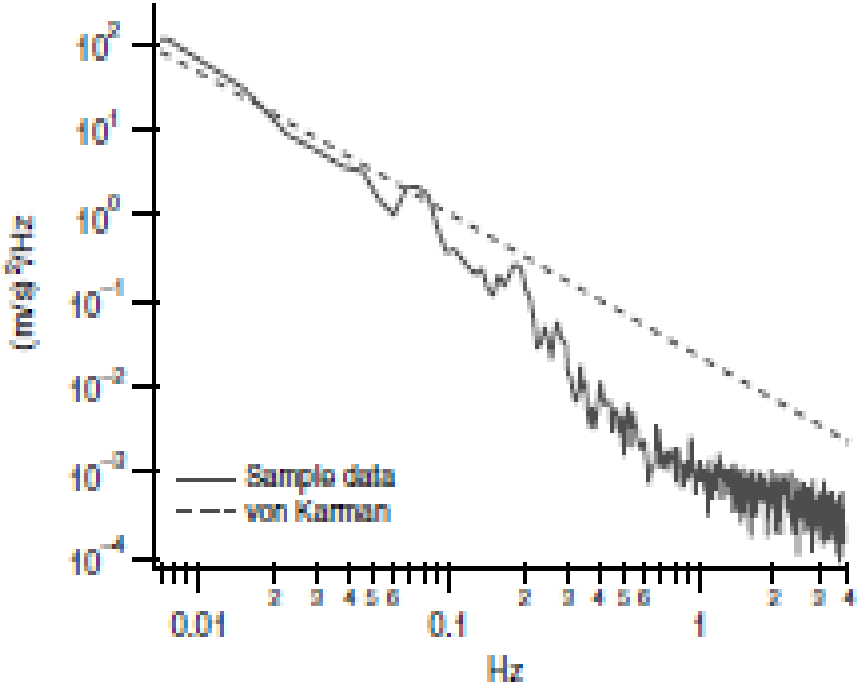
- Wind velocity is measured based on the time of flight of sonic pulses between pairs of transducers.
- The spatial resolution is determined by the path length between transducers (typically 10 to 20 cm).
- Sonic anemometers can be used for turbulence measurements with fine temporal resolution (20 Hz or better).
- A summary review of the state of the art of sonic anemometry in wind engineering applications is given by Pedersen et al. (2006).

Acoustic Doppler Sensors (SODAR)

- SODAR (standing for SOund Detection And Ranging) is classified as a remote sensing system, since it can make measurements without placing an active sensor at the point of measurement.
- Since such devices do not need tall (and expensive) towers for their use, the potential advantages of their use are obvious. Remote sensing is used extensively for meteorological
- and aerospace purposes, but only in recent times has it been used for wind siting and performance measurements.
- SODAR is based on the principle of acoustic backscattering. In order to measure the wind profile with SODAR, acoustic pulses are sent vertically and at a small angle to the vertical.
- For measurement of three-dimensional wind velocity, at least three beams in different directions are needed.

Wind Speed Probability Density Functions

- The probability density function that best describes this type of behavior for turbulence is the Gaussian, or normal, distribution.



$$S(f) = \frac{\sigma_u^2 4(L/U)}{\left[1 + 70.8(fL/U)^2\right]^{5/6}}$$

Figure 2.17 Wind data power spectral density functions

Weibull distribution

- Use of the Weibull probability density function requires knowledge of two parameters: k, a shape factor, and c, a scale factor.
- Both of these parameters are functions of U and sU.
- The Weibull probability density function and the cumulative distribution function are given by:
- As the value of k increases, the curve has a sharper peak, indicating that there is less wind speed variation.

$$p(U) = \left(\frac{k}{c}\right)\left(\frac{U}{c}\right)^{k-1} \exp\left[-\left(\frac{U}{c}\right)^{k}\right]$$

$$F(U) = 1 - \exp\left[-\left(\frac{U}{c}\right)^{k}\right]$$

Rayleigh Distribution

- This is the simplest velocity probability distribution to represent the wind resource since it requires only a knowledge of the mean wind speed, U.
- The probability density function and the cumulative distribution function are given by:
- A larger value of the mean wind speed gives a higher probability of higher wind speeds.

$$p(U) = \frac{\pi}{2}\left(\frac{U}{\overline{U}^2}\right)\exp\left[-\frac{\pi}{4}\left(\frac{U}{\overline{U}}\right)^2\right]$$

$$F(U) = 1-\exp\left[-\frac{\pi}{4}\left(\frac{U}{\overline{U}}\right)^2\right]$$

$$p(u) = \frac{1}{\sigma_u \sqrt{2\pi}} \exp\left[-\frac{(u-U)^2}{2\sigma_u^2}\right]$$

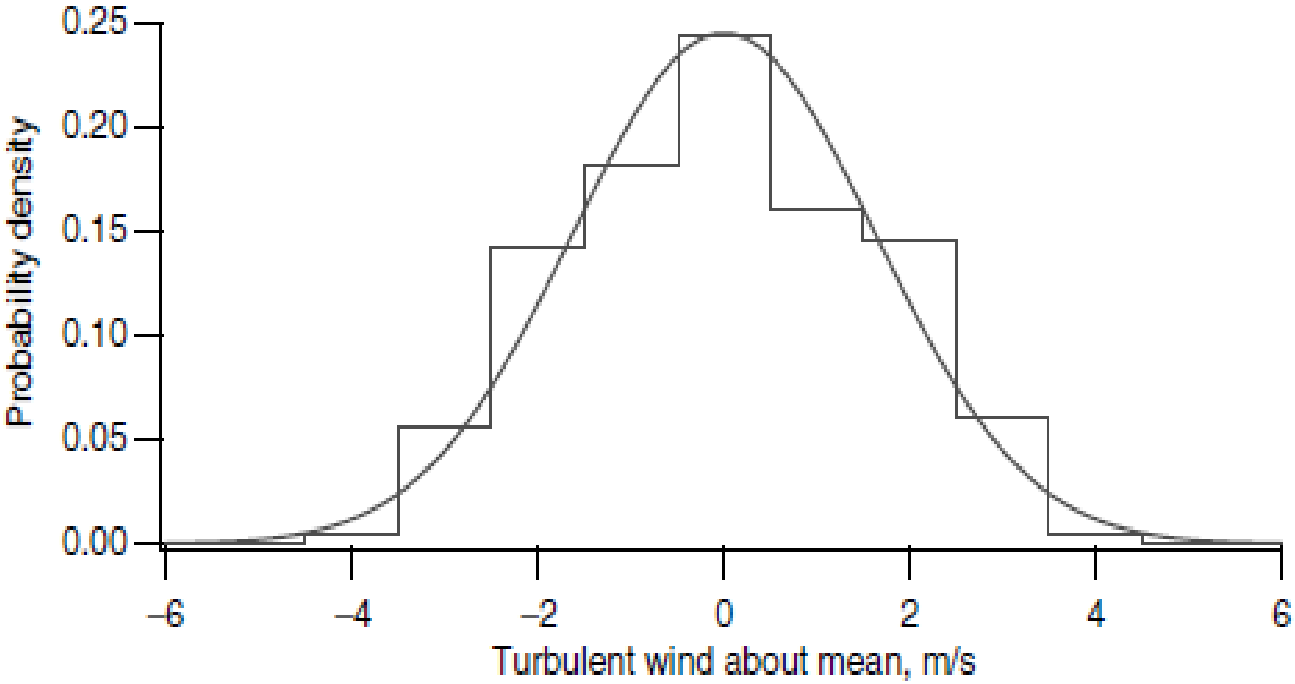


Figure 2.15 Gaussian probability density function and histogram of wind data

- https://www.youtube.com/watch?v=qSWm_nprfqE
- https://www.youtube.com/watch?v=LDRHZ5ry-Ts
- https://www.youtube.com/watch?v=ecT4HvZnaLE
- http://xn--drmstrre-64ad.dk/wp-content/wind/miller/windpower%20web/en/tour/wtrb/yaw.htm

WIND TURBINES

A wind turbine is a machine for converting the kinetic energy in wind into mechanical energy.

WINDMILLS

If the mechanical energy is used directly by machinery, such as a pump or grinding stones, the machine is usually called a windmill.

WIND TURBINES

If the mechanical energy is then converted to electricity, the machine is called a wind generator.

Types of Wind Turbines

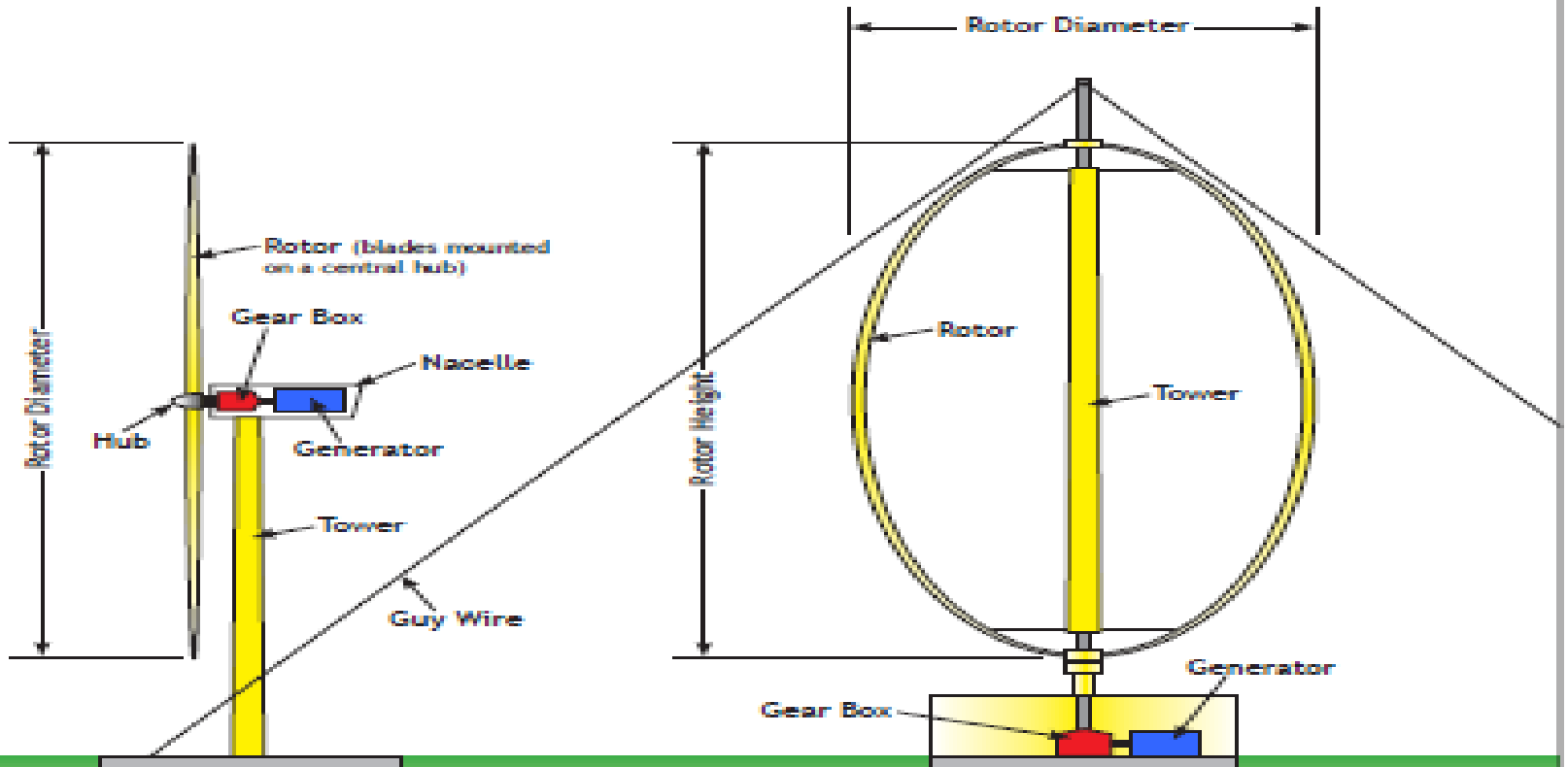
Wind turbines are classified into two general types: horizontal axis and vertical axis. A horizontal axis machine has its blades rotating on an axis parallel to the ground. A vertical axis machine has its blades rotating on an axis perpendicular to the ground. There are a number of available designs for both and each type has certain advantages and disadvantages. However, compared with the horizontal axis type, very few vertical axis machines are available commercially.

Horizontal Axis Wind Turbine (HAWT)

Vertical Axis Wind Turbine (HAWT)

Parts of a Wind Turbine

- The nacelle contains the key components of the wind turbine, including the gearbox, and the electrical generator.
- The tower of the wind turbine carries the nacelle and the rotor. Generally, it is an advantage to have a high tower, since wind speeds increase farther away from the ground.
- The rotor blades capture wind energy and transfer its power to the rotor hub.
- The generator converts the mechanical energy of the rotating shaft to electrical energy
- The gearbox increases the rotational speed of the shaft for the generator.



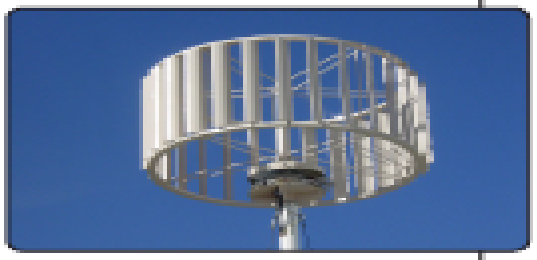
HAWT
(Horizontal Axis Wind Turbine)

VAWT
(Vertical Axis Wind Turbine)

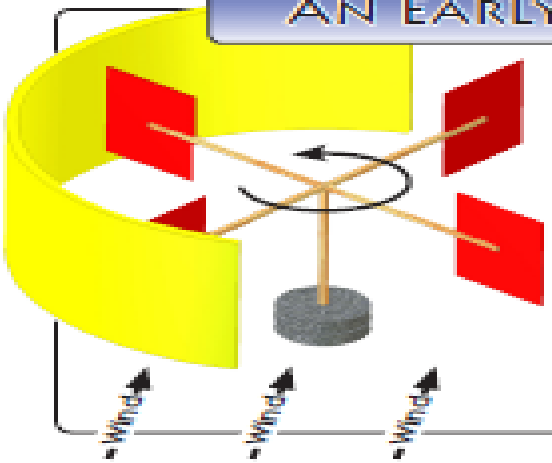
TeacherGeek 2006

Vertical Axis Wind Turbines

Although vertical axis wind turbines have existed for centuries, they are not as common as their horizontal counterparts. The main reason for this is that they do not take advantage of the higher wind speeds at higher elevations above the ground as well as horizontal axis turbines.



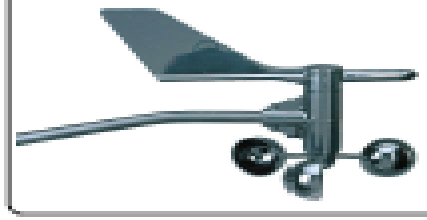
An Early Wind Turbine

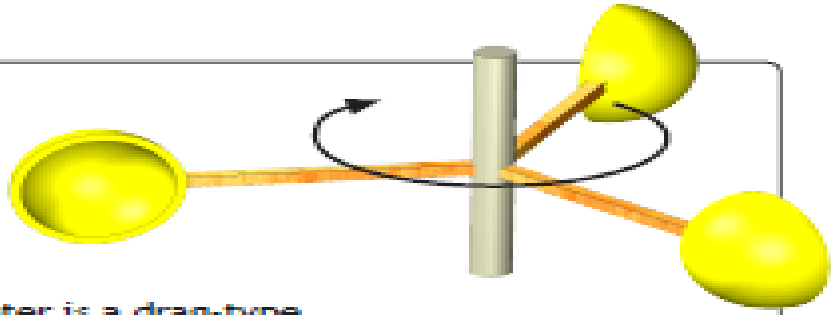


The Persian windmill was used around 1000 b.c. to turn a grindstone. It is the oldest known windmill design. The machine works by blocking the wind blowing on ½ of its sails. The sails exposed to the wind are pushed downwind due to drag, causing the windmill to rotate.

Anonometer

The anonometer is an instrument for measuring the speed of airflow.

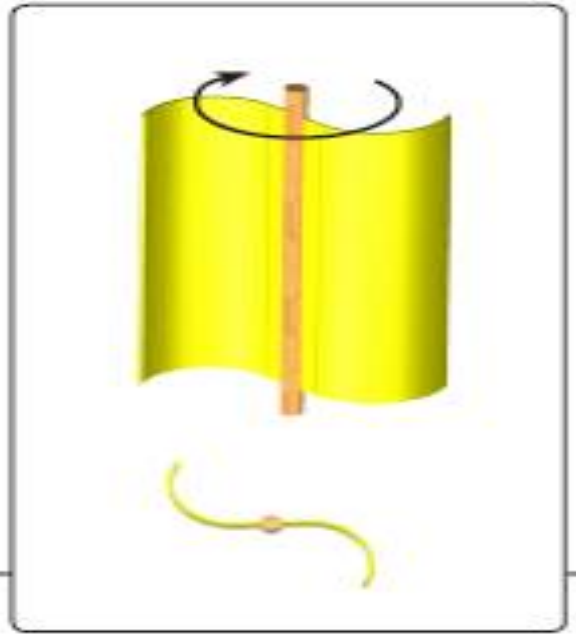




A cup anemometer is a drag-type vertical axis wind turbine

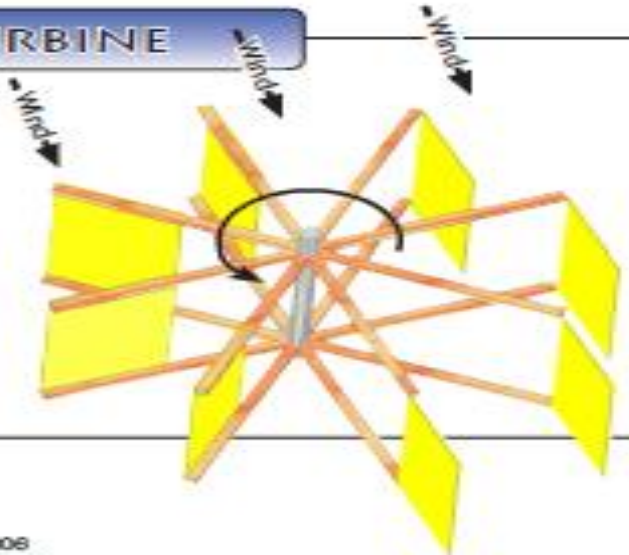
Savonius Wind Turbine

The Savonius turbine is S-shaped if viewed from above. This drag-type VAWT turns relatively slowly, but yields a high torque. It is useful for grinding grain, pumping water, and many other tasks, but its slow rotational speeds make it unsuitable for generating electricity on a large-scale.



Flapping Panel Wind Turbine

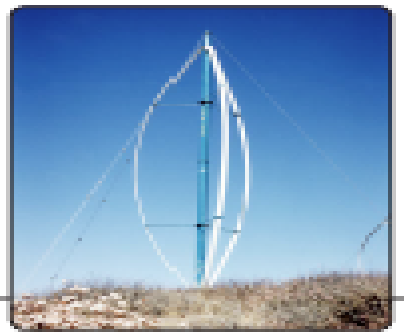
This illustration shows the wind coming from one direction, but the wind can actually come from any direction and the wind turbine will work the same way.

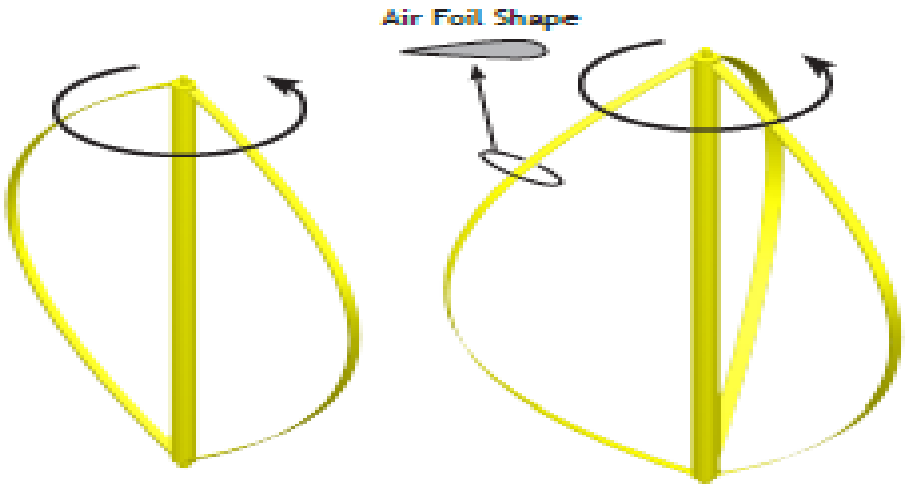


TeacherGeek 2008

Darrieus Wind Turbine

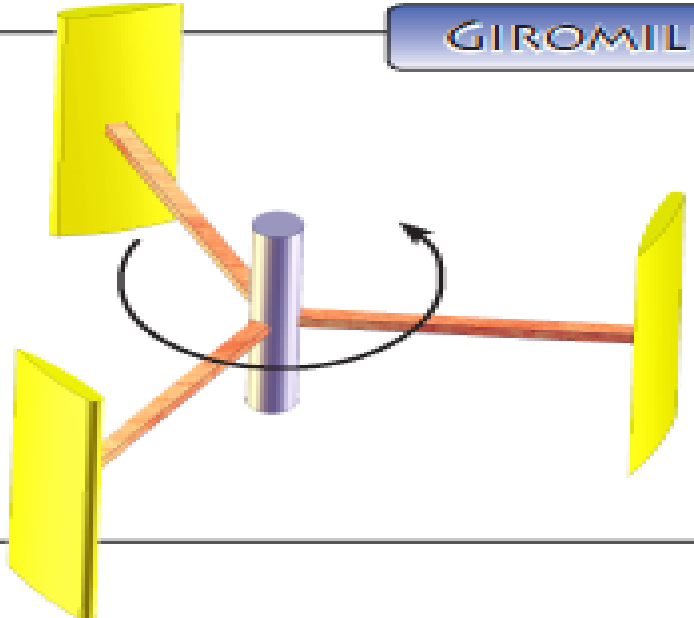
The Darrieus turbine is the most famous vertical axis wind turbone. It is characterised by its C-shaped rotor blades which give it its eggbeater appearance. It is normally built with two or three blades.





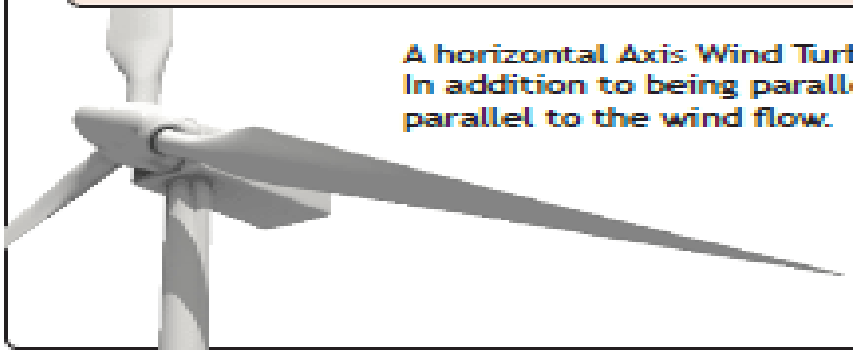
The Darrieus turbine is not self starting. It needs to start turbing before the wind will begin rotating it.

Giromill Wind Turbine



The giromill is typically powered by two or three vertical aerofoils attached to the central mast by horizontal supports. Giromill turbines work well in turbulent wind conditions and are an affordable option where a standard horizontal axis windmill type turbine is unsuitable.

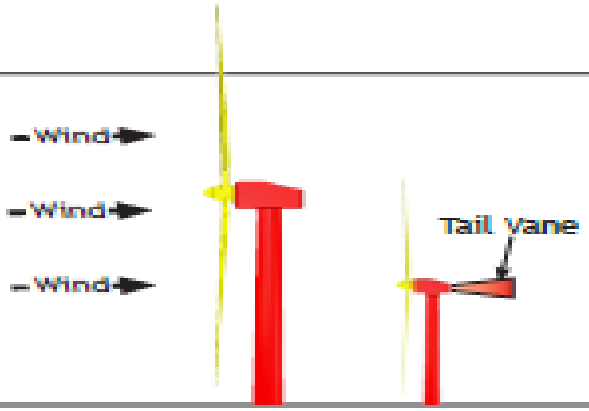
A. Horizontal Axis Wind Turbines



A horizontal Axis Wind Turbine is the most common wind turbine design. In addition to being parallel to the ground, the axis of blade rotation is parallel to the wind flow.

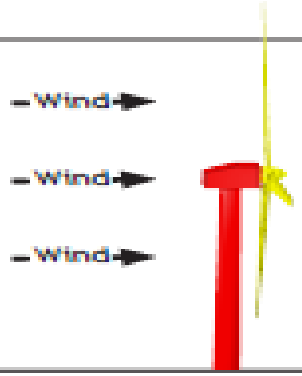
Up-Wind Turbines

Some wind turbines are designed to operate in an upwind mode (with the blades upwind of the tower). Large wind turbines use a motor-driven mechanism that turns the machine in response to a wind direction. Smaller wind turbines use a tail vane to keep the blades facing into the wind.



Down-Wind Turbines

Other wind turbines operate in a downwind mode so that the wind passes the tower before striking the blades. Without a tail vane, the machine rotor naturally tracks the wind in a downwind mode.



HAWT

Advantages

• Blades are to the side of the turbines center of gravity, helping stability

• Ability to wing warp, which gives the turbine blades the best angle of attack

• Ability to pitch the rotor blades in a storm to minimize damage

• Tall tower allows access to stronger wind in sites with wind shear

• Tall tower allows placement on uneven land or in offshore locations

• Can be sited in forest above tree-line

• Most are self-starting

Disadvantages

• Difficulty operating in near ground winds

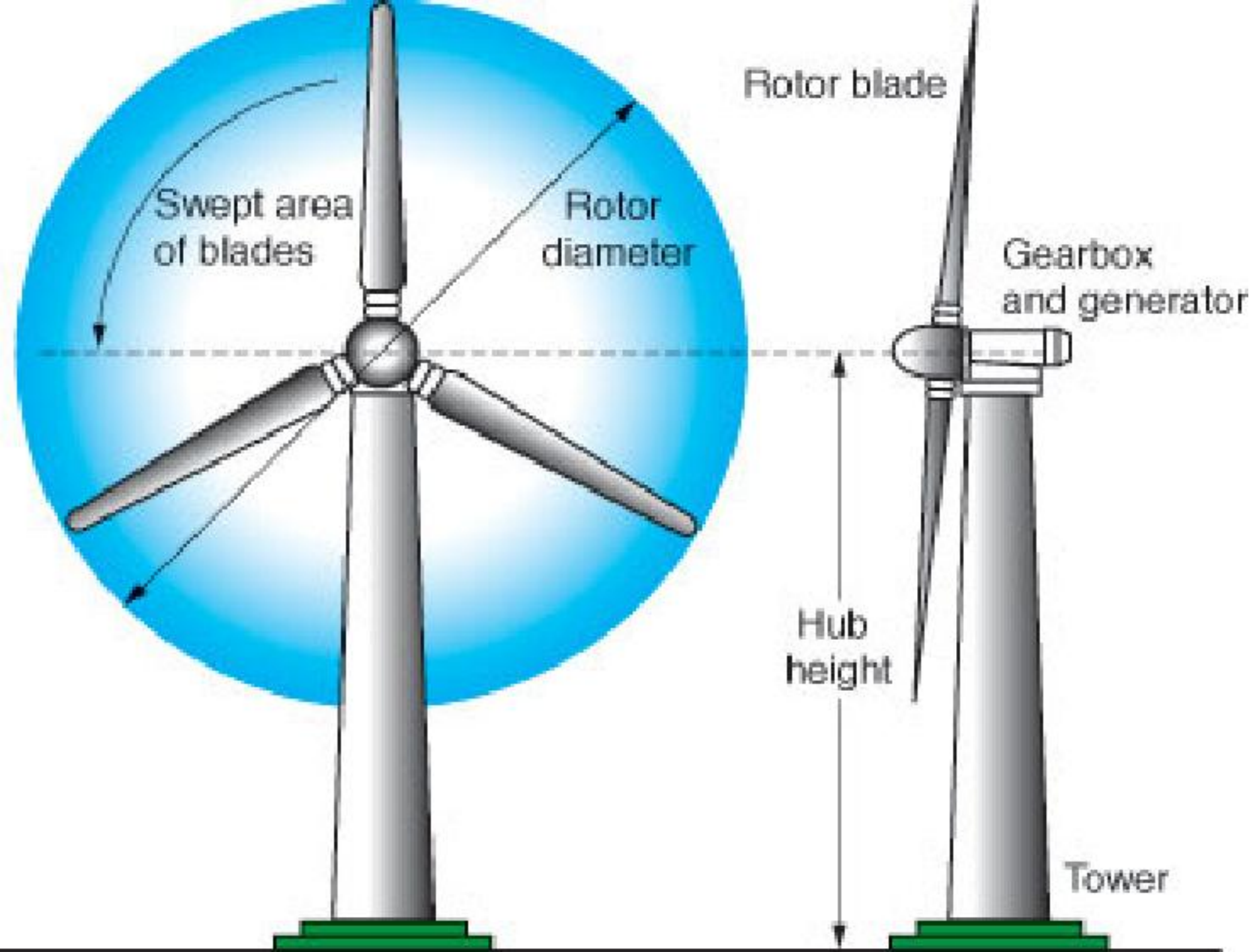
• Difficult to transport (20% of equipment costs)

• Difficult to install (require tall cranes and skilled operators)

• Effect radar in proximity

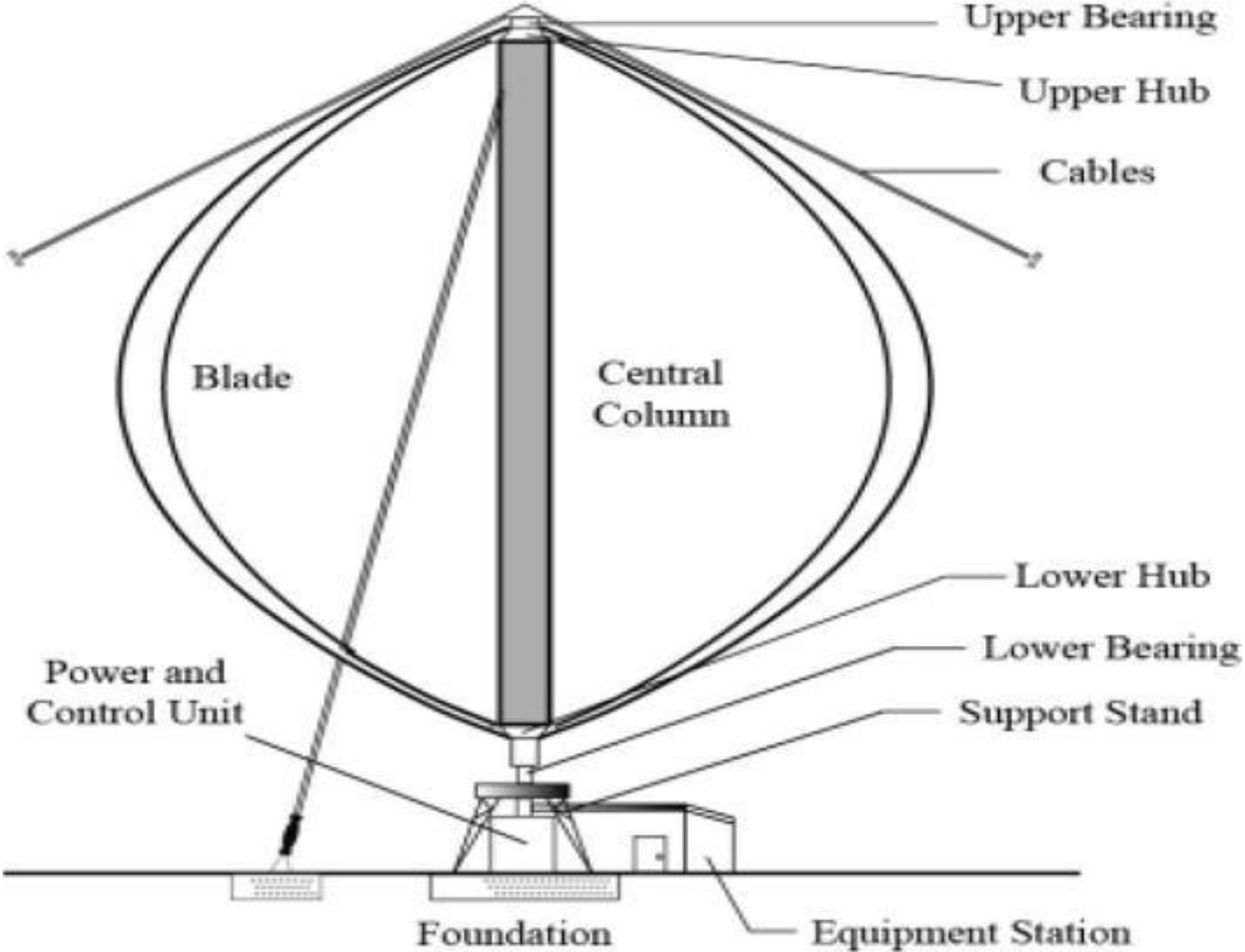
• Local opposition to aesthetics

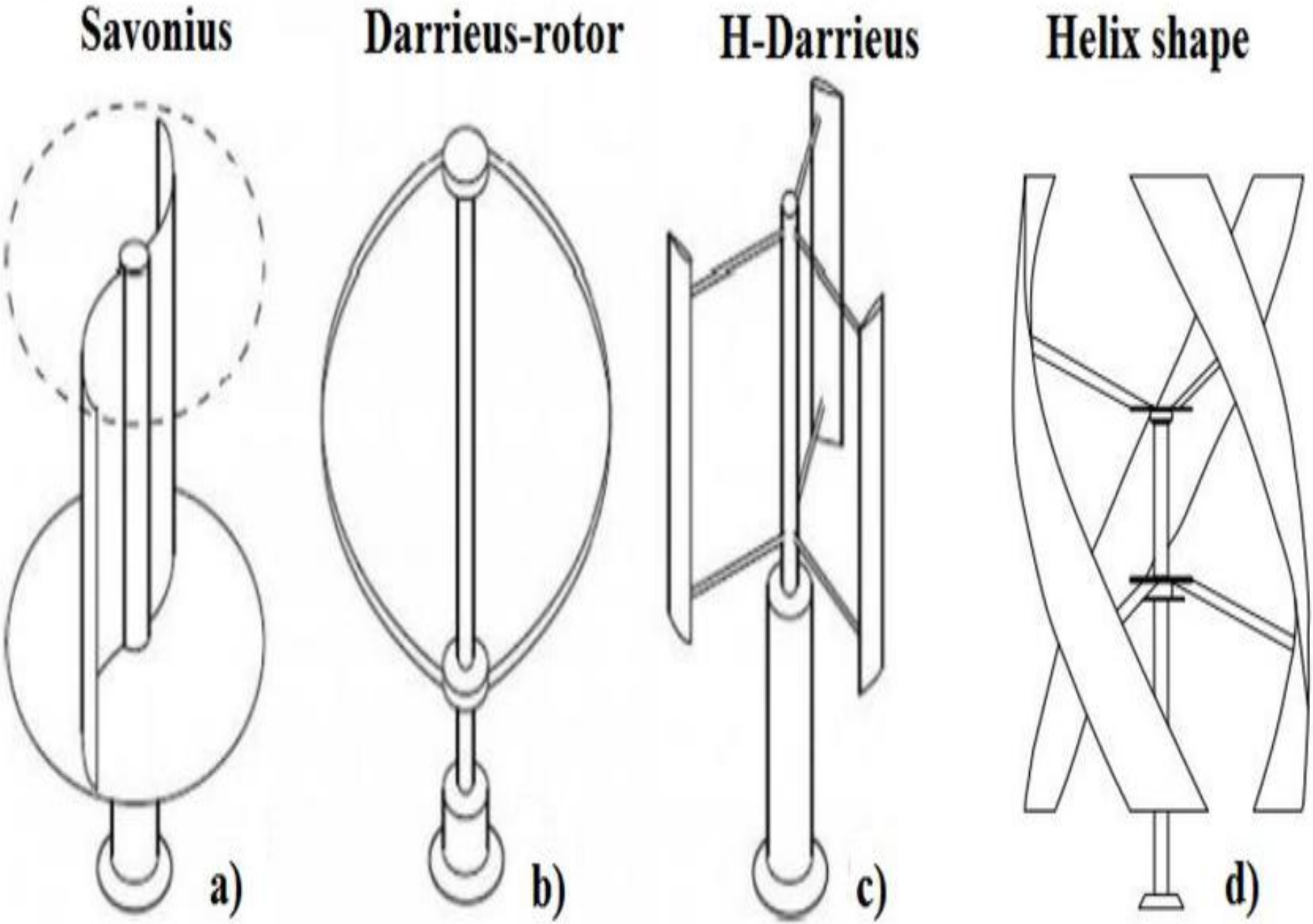
• Difficult maintenance


Swept area
of blades
Rotor
diameter
Rotor blade
Gearbox
and generator
Hub
height
Tower

VAWT

- **Advantages**
- • Easy to maintain
- • Lower construction and transportation costs
- • Not directional
- • Most effective at mesas, hilltops, ridgelines and passes
- **Disadvantages**
- • Blades constantly spinning back into the wind causing drag
- • Less efficient
- • Operate in lower, more turbulent wind
- • Low starting torque and may require energy to start turning


Upper Bearing
Upper Hub
Cables
Blade
Central
Column
Lower Hub
Lower Bearing
Power and
Control Unit
Support Stand
Foundation
Equipment Station


Savonius
Darrieus-rotor
H-Darrieus
Helix shape
a)
b)
c)
d)

Ducted Wind Turbine

Advantages

• Less visual impact on buildings architecture than traditional HAWT or VAWT turbines

• Make use of unused roof space in cities

• Allows energy need to be met on-site avoiding transmission losses associated with centralized energy generation

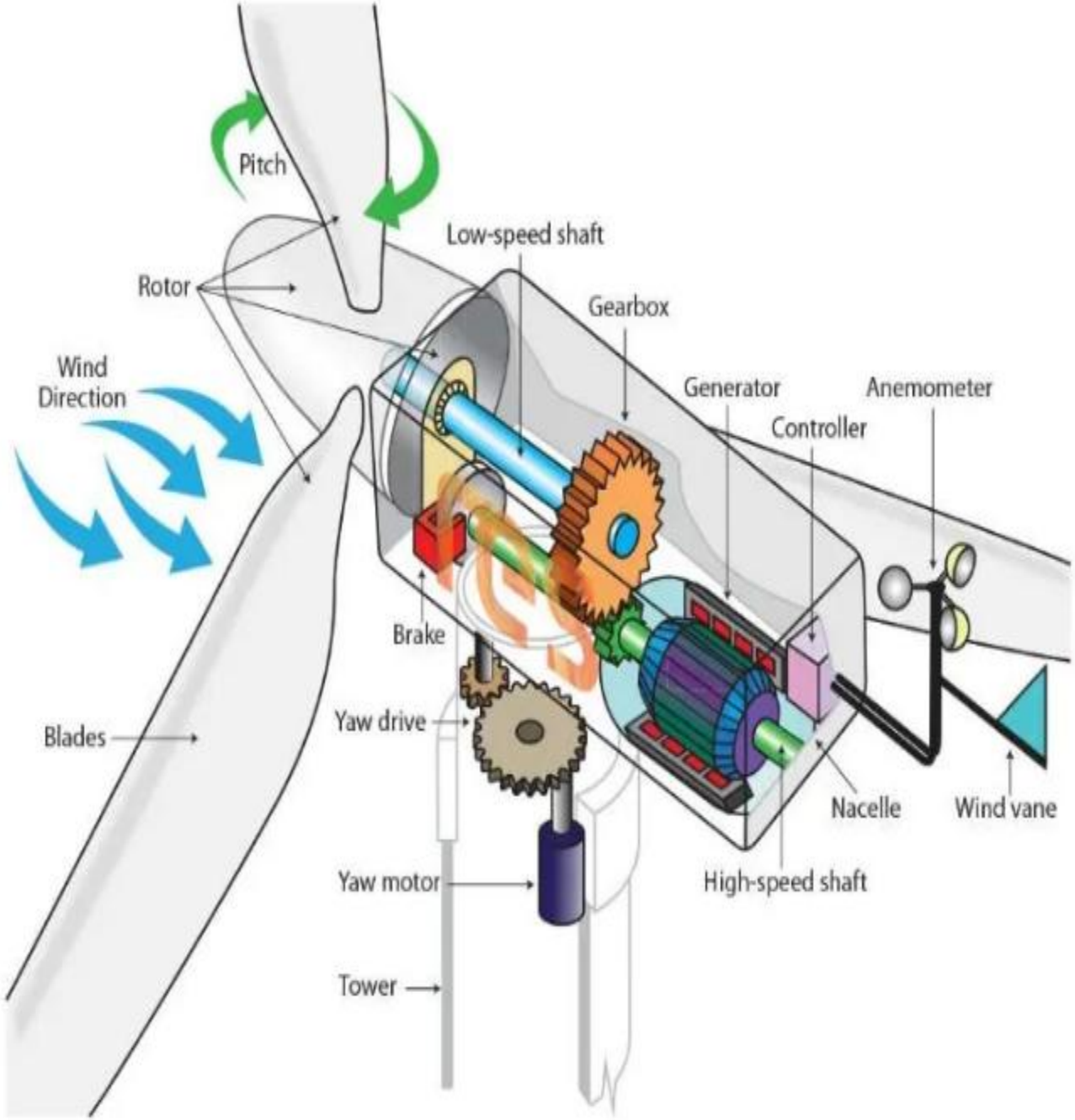
Disadvantages

• Suitable for urban environments, but not households (only effective on urban high-rise buildings)

• Uni-directional. Fixed position and are dependent upon wind blowing in the correct direction

• Much more research and development is needed. Research in this field is growing as people become more interested in urban wind generation.

• Research has to be done to determine energy production potential

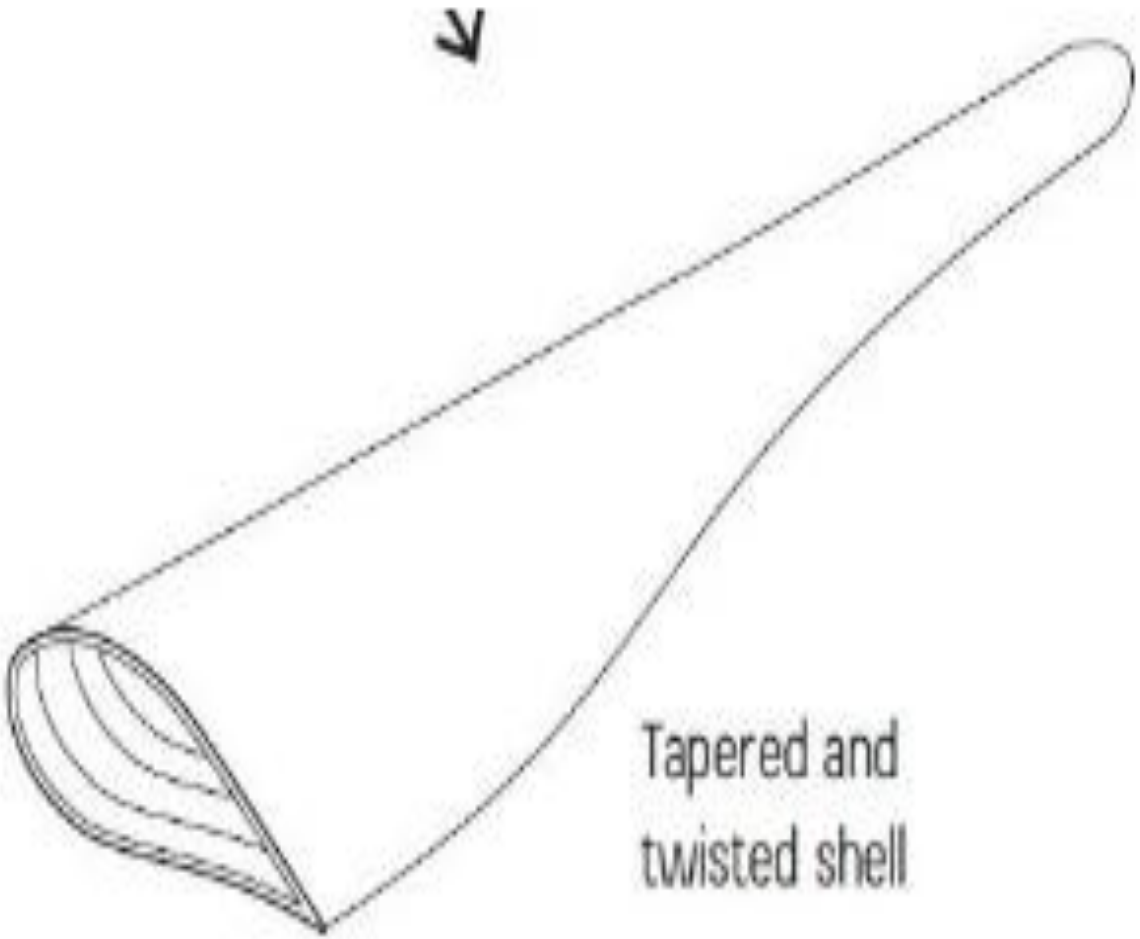

Pitch
Low-speed shaft
Rotor
Gearbox
Wind
Direction
Generator
Anemometer
Controller
Brake
Yaw drive
Blades
Nacelle
Wind vane
Yaw motor
High-speed shaft
Tower

Activate

Wind Turbine Subsystems

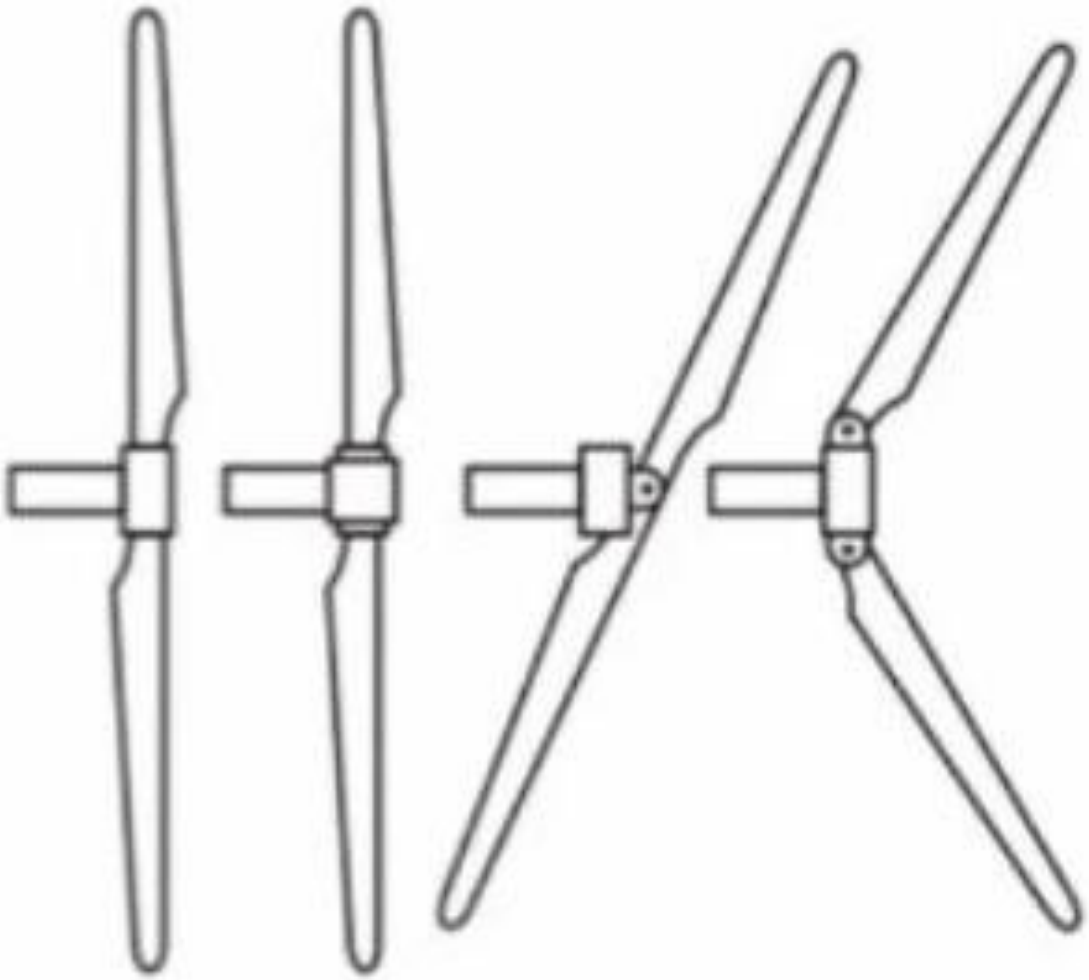
- The major wind turbine subsystem are following
- **Rotor:** Blades and Hub
- **Drive Train:** Low-Speed Shaft (LSS), Bearings, Couplings, Gear Box, High-Speed Shaft (HSS), Brakes
- **Electrical:** Generator, Power Electronics
- **Control:** Pitch motor and gears, Yaw motor, gears and brakes, sensors (wind and direction)
- **Support Structures:** Tower, Nacelle

- **Rotor**
- The blades and the hub together are called the rotor.
- **Blades**
- Blades can be pitched and can have control surfaces (flaps).
- Blades can be twisted, tapered, and coned.
- Blade materials are Wood and cloth, Glass (reinforced) fiber polyester (GFP), Glass fiber epoxy (GFE), Carbon fiber epoxy, Wood epoxy, Aluminum, Steel (heavy)



Wind Turbine blade

- **Hub**
- Blades attached to the Hub. Hub options (from left to right)
- Rigid
- Rigid/pitching
- Teetering
- Hinged hub



Wind Turbine Hub

Drive Train

Drive Train consists of the Low-Speed Shaft (LSS), Bearings, Couplings, Gear Box, High-Speed Shaft (HSS), and Brakes.

- **Low-Speed Shaft**
- The low-speed shaft is also called the main shaft.
- Rotor RPM is low (30-60 RPM) and the torque is high.
- As the rotor size increases, the torque and moments correspondingly increase.
- The torque transmission path is identified as the “load-path”.
- The main/low-speed shaft is usually made of steel and must be able to carry very large torque loading.
- It transfers the torque from the rotor to the rest of the drive train, while also supporting the weight of the rotor.
- The main shaft is supported by bearings which transfer the reactionary loads to the main frame of the turbine.

GearBox

- The gearbox connects the low-speed shaft to the high-speed shaft. It increases the rotational speeds from about 30 to 60 rotations per minute (RPM) to about 1500 to 1800 RPM, the rotational speed required by most generators to produce electricity.
- The gearbox is a costly (and heavy) part of the wind turbine. Engineers are exploring “direct-drive” generators that operate at lower rotational speeds and do not need gearboxes.
- Gearboxes in wind power transform slow speed, high torque wind turbine rotation to higher speed required by the generator, which converts the mechanical power to electricity.
- **Choices:**
- Planetary or parallel
- Shaft bearing and location
- Number of stages
- Speed-up ratio

Gear system

Big gear
Rolling contact bearing
Big gear shaft
Pinion shaft
Rolling contact bearing
Pinion
Bearing Pedestal
Lower box
www.pengky.cn

Coupling

- Couplings connect the shafts together and transmit torque between the two shafts.
- Couplings can be effectively used to dampen torque fluctuations in the main shaft before power is converted to electricity.
- They are typically found between the main shaft and the gearbox, and between the gearbox output and the generator.

Brakes

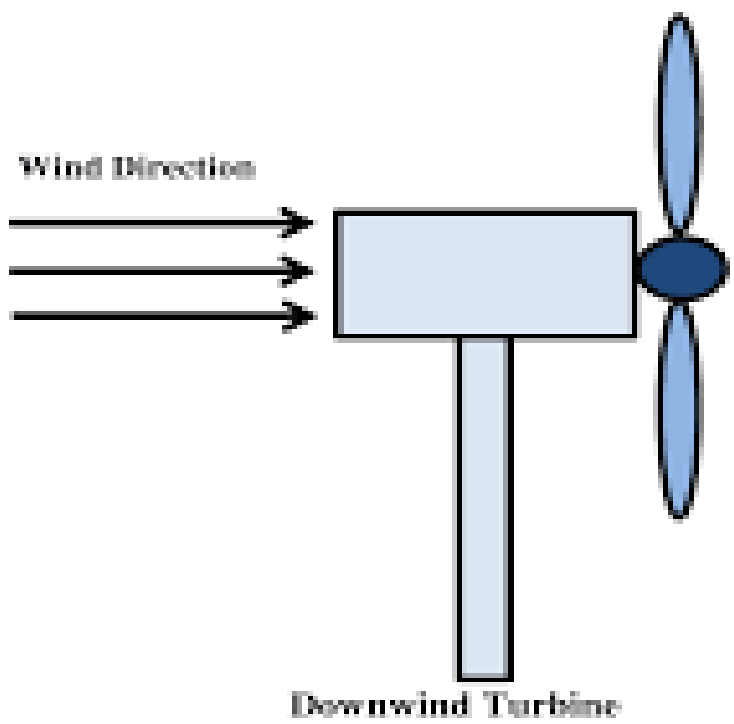
- Turbines do not operate above certain wind speeds because they might be damaged by the high winds and associated loads.
- Brakes – A disc brake can be applied mechanically, electrically, or hydraulically to stop the rotor in emergencies, for maintenance, and when the threshold speed is exceeded.

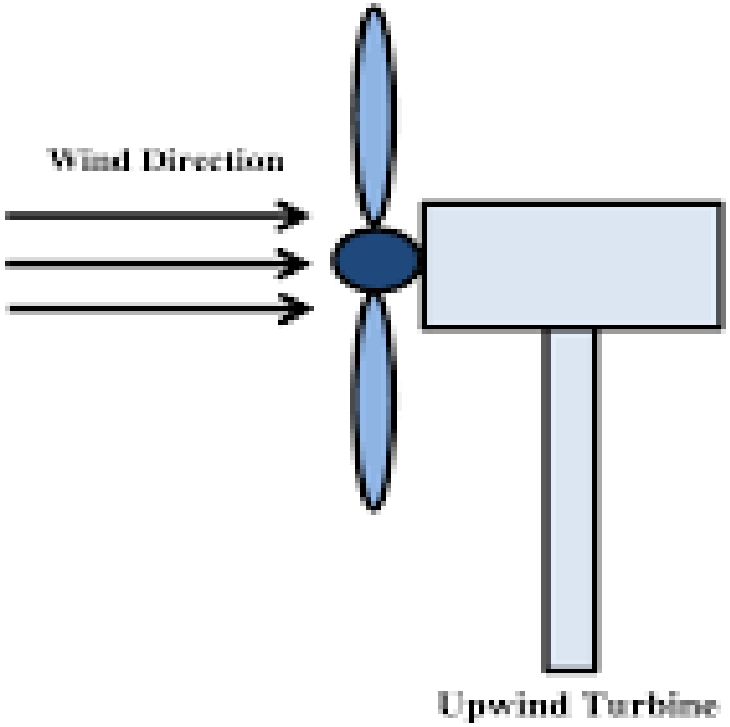
Yaw System

- **Free yaw** (typical in downwind/smaller turbines).
- The turbine automatically orients due to the aerodynamic loads on the blades.
- **Utility-grade turbines** employ a yaw drive (gear-motor) and direction sensor (wind vane) to orient the rotor blades into the wind.
- The difference between the orientation of the rotor and the direction of the wind is used to activate the yaw motion.
- A yaw motor, pinion gear, bull gear, and yaw brakes make up the yaw system.
- **Yaw motor** – Powers the yaw drive.
- **Yaw drive** – Upwind turbines face into the wind; the yaw drive is used to keep the rotor facing into the wind as the wind direction changes.
- **Downwind turbines** do not require a yaw drive, the wind blows the rotor downwind.

Upwind and down wind

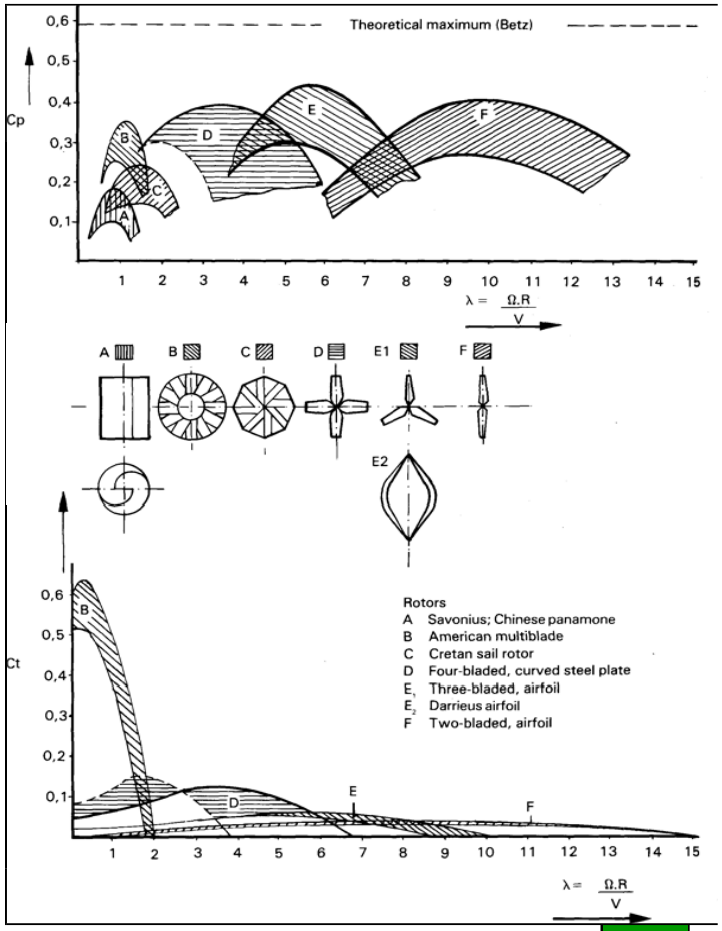
- 'Upwind' is in the direction towards the source of the wind, and 'Downwind' is in the direction away from the source of the wind





Nacelle, Rotor & Hub

- Main Rotor Design Method (ideal case):
  1. Determine basic configuration: orientation and blade number
  2. take site wind speed and desired power output
  3. Calculate rotor diameter (accounting for efficiency losses)
  4. Select tip-speed ratio (higher → more complex airfoils, noise) and blade number (higher efficiency with more blades)
  5. Design blade including angle of attack, lift and drag characteristics
  6. Combine with theory or empirical methods to determine optimum blade shape



Graphic source Wind power: http://www.fao.org/docrep/010/ah810e/AH810E10.htm

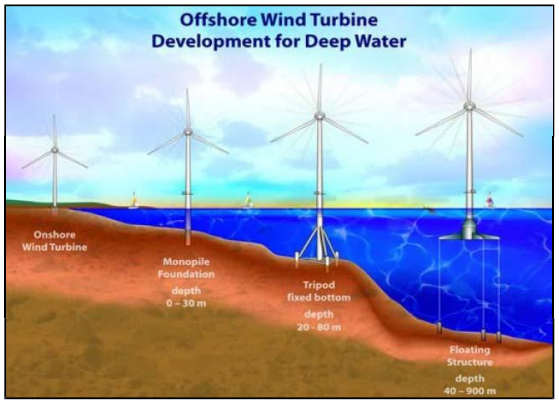
Nacelle

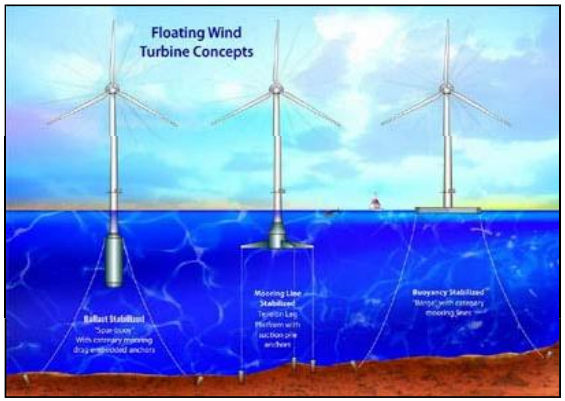
- The nacelle houses all the major components of a modern wind energy conversion system, except the rotor. The nacelle sits atop the tower.
- It is more of a protective cover for the components and is made of lightweight material.
- The nacelle does not carry any load and thus is not a structural member.

Foundations and Tower

- Evolution from truss (early 1970s) to monopole towers

- Many different configurations proposed for offshore

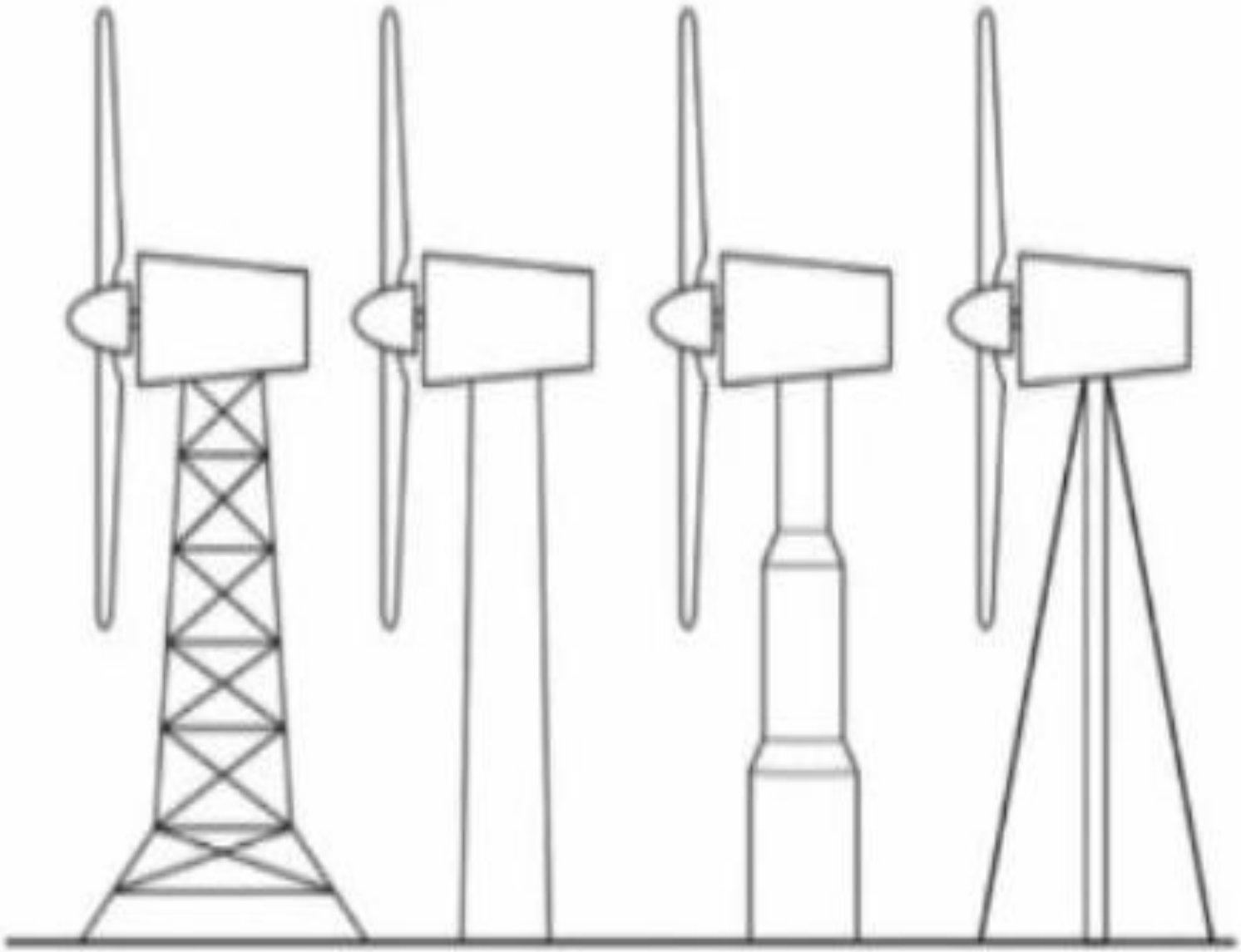




Images from National Renewable Energy Laboratory

Towers

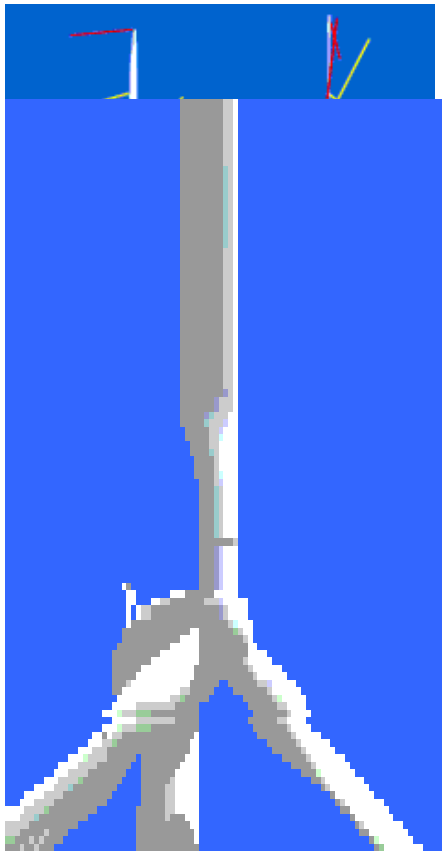
- Towers are generally made of galvanized steel (tubular) or lattice (truss). Portions may be made out of concrete as well.
- Smaller turbines can be supported with guy lines.
- The tower supports the nacelle and transmits static, and dynamic forces and moments to the foundation.
- Tower heights are usually of the same order as the rotor diameter.
- For modern turbines, the tower is the most expensive part.
- Tubular towers provide protected access to components in the nacelle.



Wind Turbine Tower

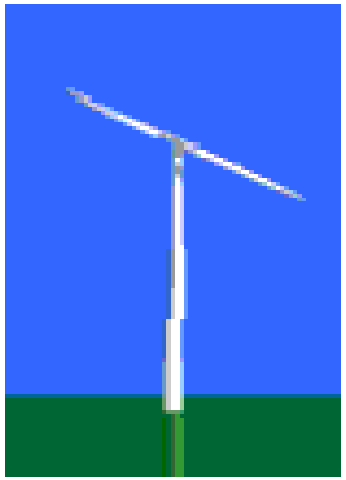
Wind Turbine Blades

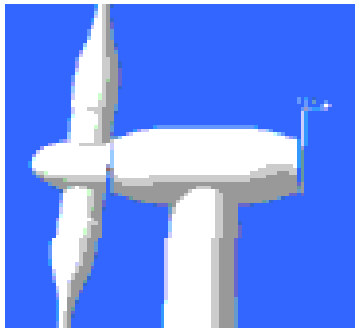
- Blade tip speed:



- Pitch control:

- 2-Blade Systems and Teetered Hubs:

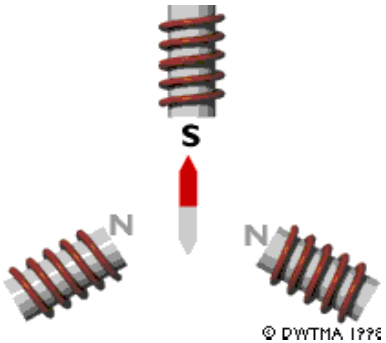


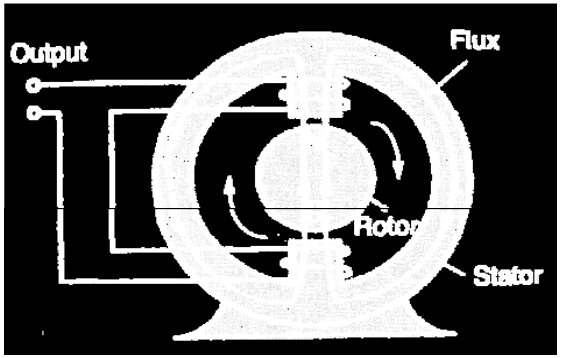


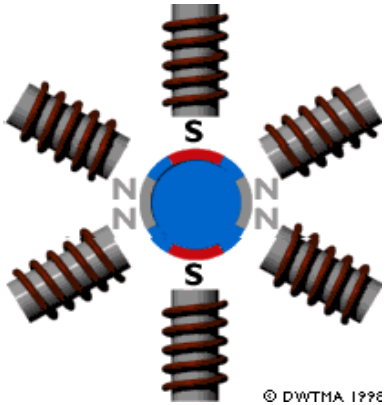
http://guidedtour.windpower.org/en/tour/wres/index.htm

Electrical Generator

- Generator:
  - Rotating magnetic field induces current







- Synchronous / Permanent Magnet Generator
  - Potential use without gearbox
  - Historically higher cost (use of rare-earth metals)
- Asynchronous / Induction Generator
  - Slip (operation above/below synchronous speed) possible
  - Reduces gearbox wear

Masters, Gilbert, *Renewable and Efficient Electric Power Systems*, Wiley-IEEE Press, 2003
http://guidedtour.windpower.org/en/tour/wtrb/genpoles.htm .

Generator

- Converts the (rotational) mechanical energy extracted into electrical energy using magnetic fields.
- Usually alternating current operated at a constant speed to produce utility-grade 60Hz AC power.
- Since induction-type generators draw large amounts of reactive power from the utility grid, compensatory capacitors are necessary.

Generator Types

- Induction generators (fixed RPM)
- Wound-rotor induction generator with variable rotor resistance (variable RPM)
- Doubly-fed induction generator (variable RPM)
- The Doubly-Fed Induction Generators (DFIG) are more commonly used today.
- Efficient over a wider range of wind speeds.
- Minimizes gust induced power spikes in addition to being able to regulate voltage.
- Permanent Magnet Direct Drive Generators

Offshore wind energy

- Advantages:
- Offshore wind speeds tend to be faster than on land.[1]
- Small increases in wind speed yield large increases in energy production: a turbine in a 15-mph wind can generate twice as much energy as a turbine in a 12-mph wind.
- Faster wind speeds offshore mean much more energy can be generated.
- Offshore wind speeds tend to be steadier than on land.[1]
- A steadier supply of wind means a more reliable source of energy.
- Many coastal areas have very high energy needs.
- Half of the United States' population lives in coastal areas,[1] with concentrations in major coastal cities.
- Building offshore wind farms in these areas can help to meet those energy needs from nearby sources.
- Offshore wind farms have many of the same advantages as land-based wind farms –they provide renewable energy; they do not consume water; they provide a domestic energy source; they create jobs; and they do not emit environmental pollutants or greenhouse gases.[2]

- Disadvantages:
- Offshore wind farms can be expensive and difficult to build and maintain. In particular:
  - It is very hard to build robust and secure wind farms in water deeper than around 200 feet (~60 m), or over half a football field's length. Although coastal waters off the east coast of the U.S. are relatively shallow, almost all of the potential wind energy resources off the west coast are in waters exceeding this depth.[3] Floating wind turbines are beginning to overcome this challenge.
  - Wave action, and even very high winds, particularly during heavy storms or hurricanes, can damage wind turbines.[1]
  - The production and installation of power cables under the seafloor to transmit electricity back to land can be very expensive.[1]
- Effects of offshore wind farms on marine animals and birds are not fully understood.[4]
- Offshore wind farms built within view of the coastline (up to 26 miles offshore, depending on viewing conditions[5]) may be unpopular among local residents, and may affect tourism and property values.[3]

Onshore wind energy

Advantages

- Normally it takes about 2-3 months before the wind turbine has paid itself back.
- This also includes the energy, which were used to produce, install, maintain and remove the wind turbine.
- Cheaper foundation
- Cheaper integration with electrical grid network

Disadvantages

- Onshore wind is an **intermittent** source of energy, as turbines cannot generate electricity on demand, but only when the wind is blowing, and at sufficient strength.
- Onshore wind turbines have been criticised for their **visual impact**. Turbines are typically more spread out than other large-scale energy infrastructure projects and so can affect a larger area.
- The **species** such as birds and bats may be affected by turbines. However, bird fatalities due to turbine collisions are relatively low compared with those from traffic.
- Turbines can contribute to **noise pollution**,

Wind Power

- Cut-in speed: the minimum wind speed at which the machine will deliver useful power.
- Rated wind speed: the wind speed at which the rated power (generally the maximum power output of the electrical generator) is reached.
- Cut-out speed: the maximum wind speed at which the turbine is allowed to deliver power (usually limited by engineering design and safety constraints).

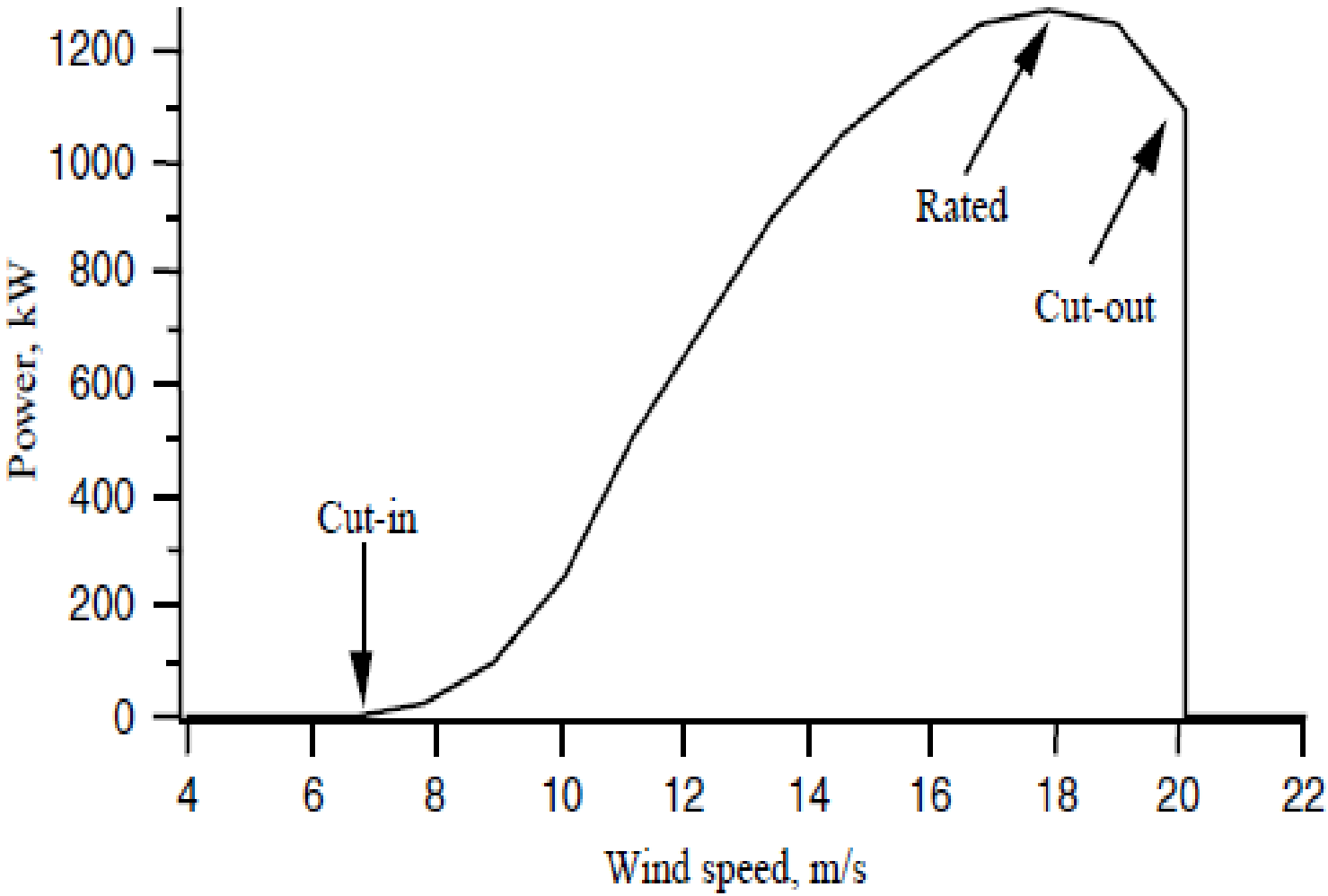


Figure 1.4 Typical wind turbine power curve

The power curve

- The power curve of a wind turbine is **a graph that indicates how large the electrical power output will be for the turbine at different wind speeds**.

Power output

- Power output is calculated as follows:
- **power = [(air density) times (swept area of blades) times (wind speed cubed)] divided by 2**.

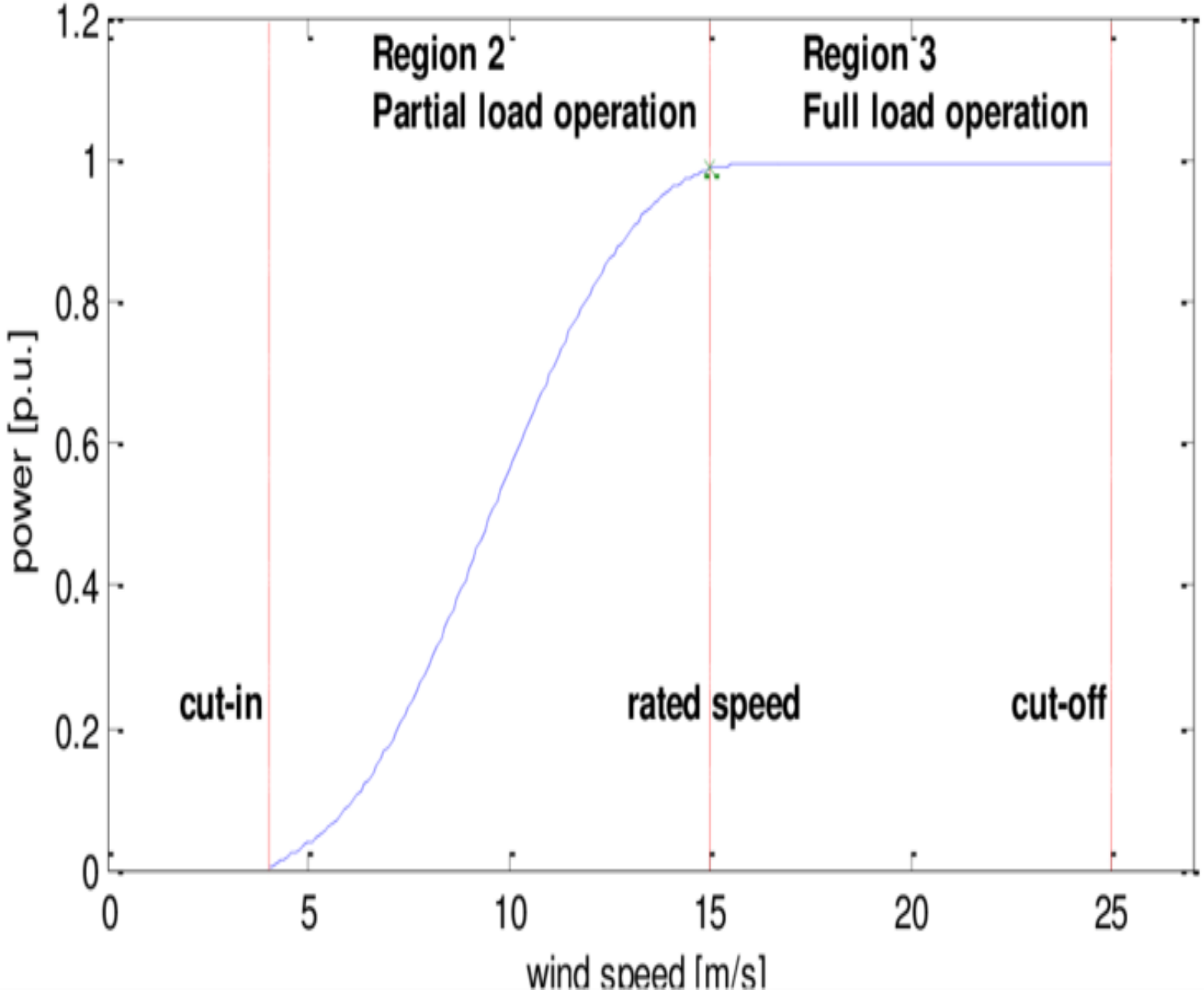
$$P = Cp\left(\frac{\rho A v^3}{2}\right), Cp = [0, 0.593]$$

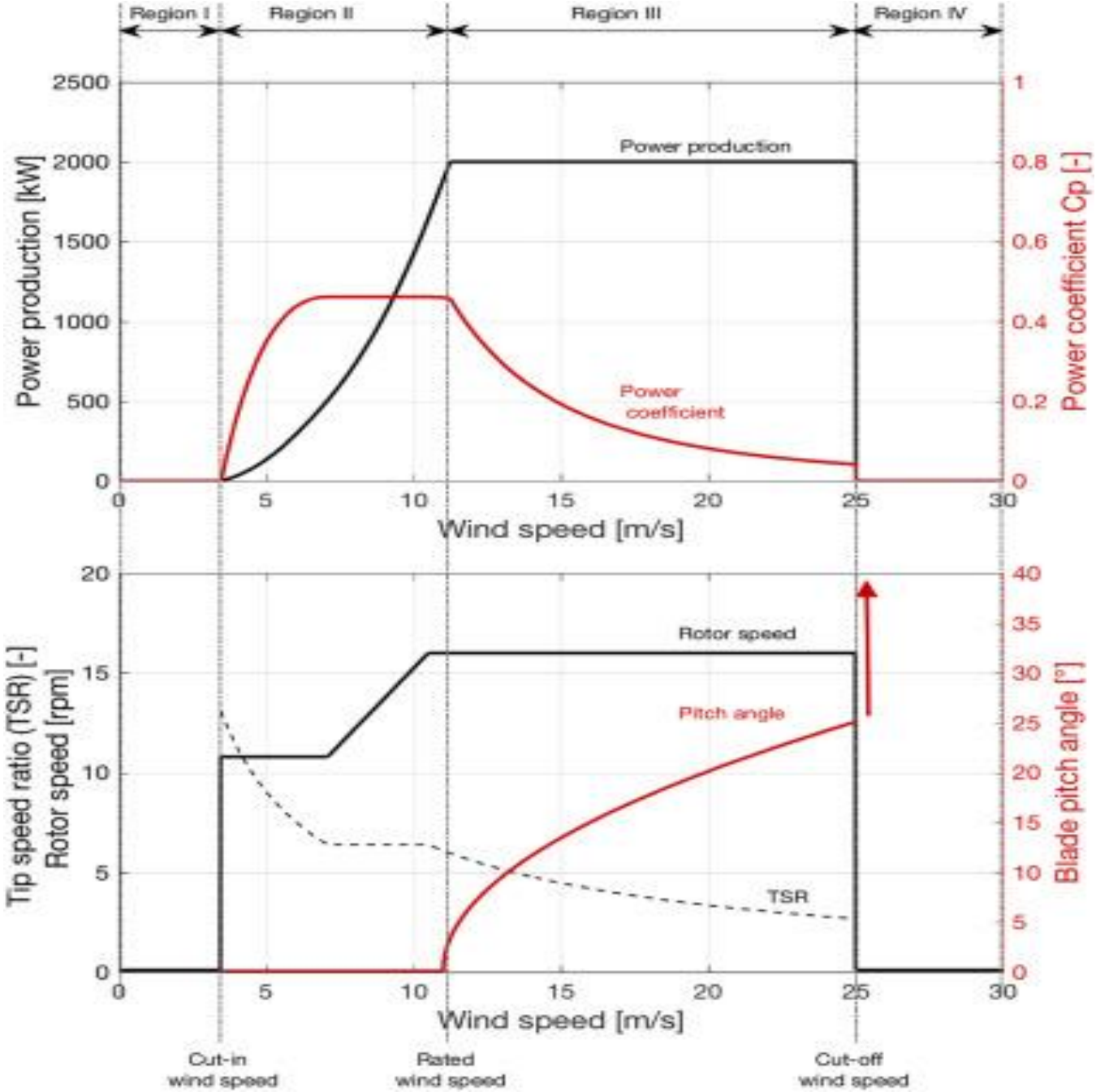
- The area is in meters squared,
- air density is in kilograms per meters cubed
- wind speed is in meters per second.

- Betz' law is a theory about the maximum possible energy to be derived from a wind turbine.
- According to Betz's law, no wind turbine of any mechanism can capture more than 16/27 (59.3%) of the kinetic energy in wind.
- It was developed in 1919 by German physicist Albert Betz.
- Betz established the ideal efficiency of this energy conversion, η, as the ratio between the ideal power transmitted to the turbine, $\eta = P_{turb} / P_{upst}$.

Speed dependency on torque

- The speed of the blade tips is ideally proportional to the speed of wind.
- The maximum torque is proportional to the speed of wind squared.
- The maximum power is proportional to the speed of wind cubed.


Region 2
Partial load operation
Region 3
Full load operation
cut-in
rated speed
cut-off
power [p.u.]
wind speed [m/s]
1.2
1
0.8
0.6
0.4
0.2
0
0
5
10
15
20
25


Region I
Region II
Region III
Region IV
Power production
Power coefficient
Power production [kW]
Power coefficient Cp [-]
Wind speed [m/s]
Rotor speed
Pitch angle
TSR
Tip speed ratio (TSR) [-]
Rotor speed [rpm]
Blade pitch angle [°]
Wind speed [m/s]
Cut-in wind speed
Rated wind speed
Cut-off wind speed

Tip Speed Ratio

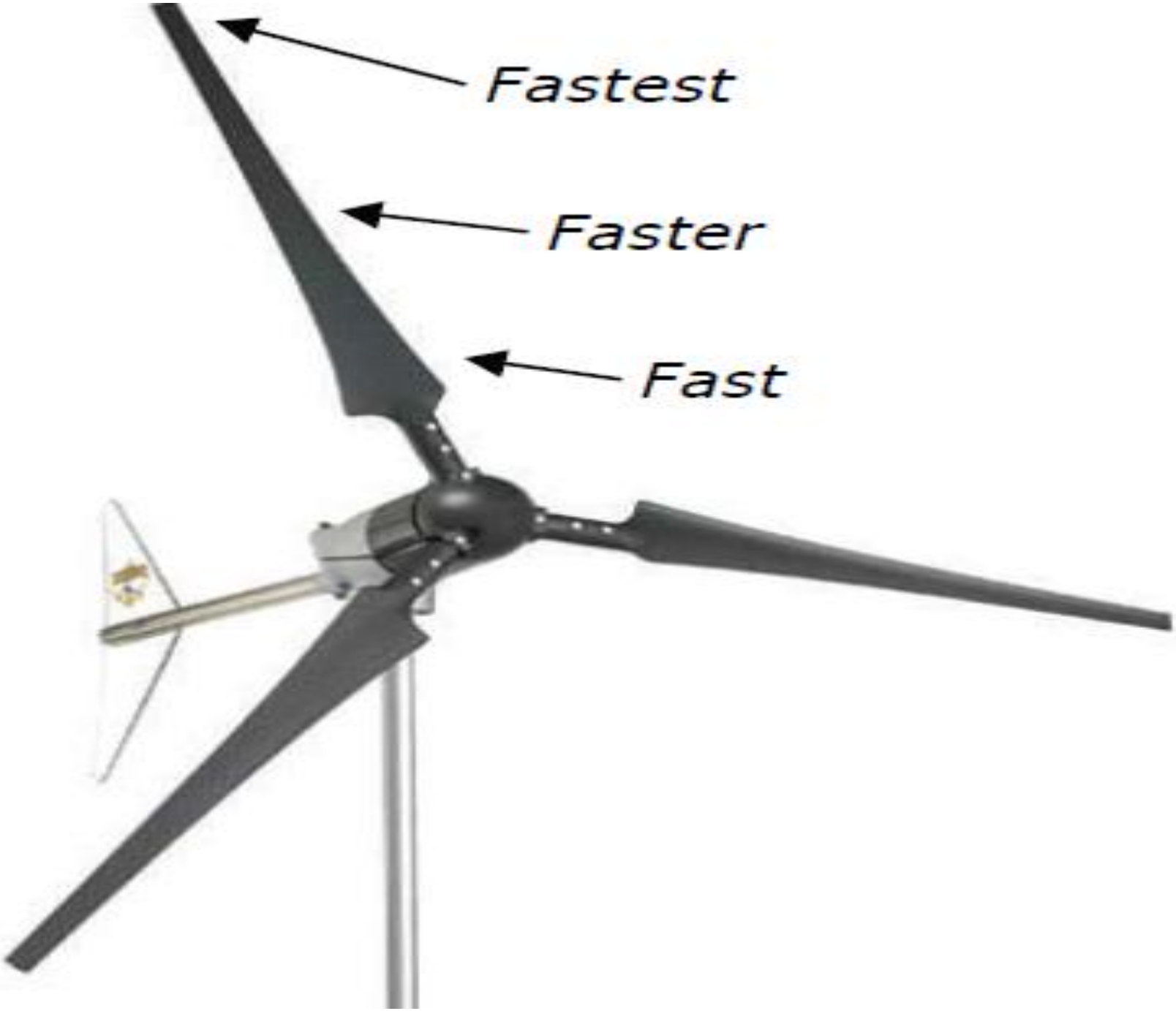
- The Tip Speed Ratio (TSR) is an extremely important factor in wind turbine design.
- **TSR refers to the ratio between the wind speed and the speed of the tips of the wind turbine blades**.
- The tip speed ratio is given **by dividing the speed of the tips of the turbine blades by the speed of the wind**
- if a 20 mph wind is blowing on a wind turbine and the tips of its blades are rotating at 80 mph, then the tip speed ration is 80/20 = 4.

How To Find The Tip Speed

- Measure the rotor radius (length of one blade)
- Speed = distance divided by time.
  - The distance travelled is the circumference ($2\prod r$).
- TSR= The blades travel one circumference ($2\prod r$) in a rotation time of T (seconds).
  - $TSR=2\pi R/T$
- 1 meter per second equals 2.237 miles per hour.

The tip speed ratio is simply a number from 0 - (theoretically a huge number, but usually 8 - 10) which tells you how fast the blades are moving in relation to the wind speed.

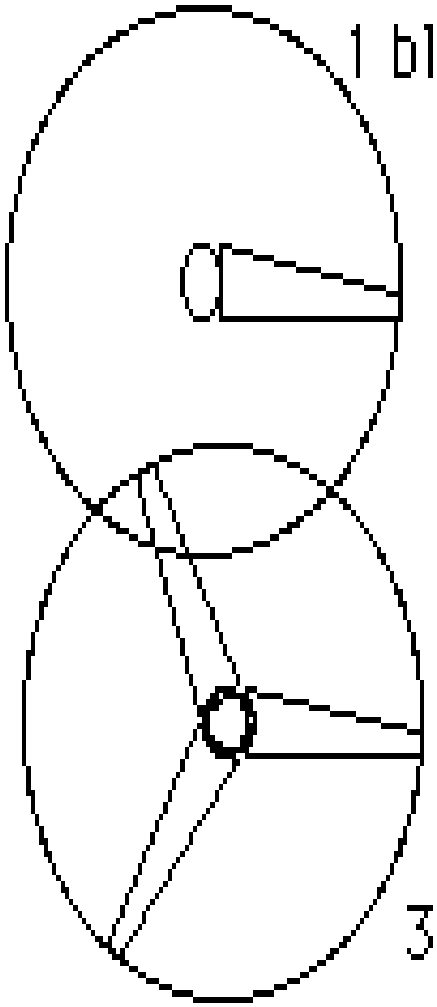
- If the rotor of the wind turbine spins too slowly, most of the wind will pass straight through the gap between the blades, therefore giving it no power!
- if the rotor spins too fast, the blades will blur and act like a solid wall to the wind. Also, rotor blades create turbulence as they spin through the air.
- If the next blade arrives too quickly, it will hit that turbulent air.
- Wind turbines must be designed with optimal tip speed ratios to get the maximum amount of power from the wind.
-


Fastest
Faster
Fast

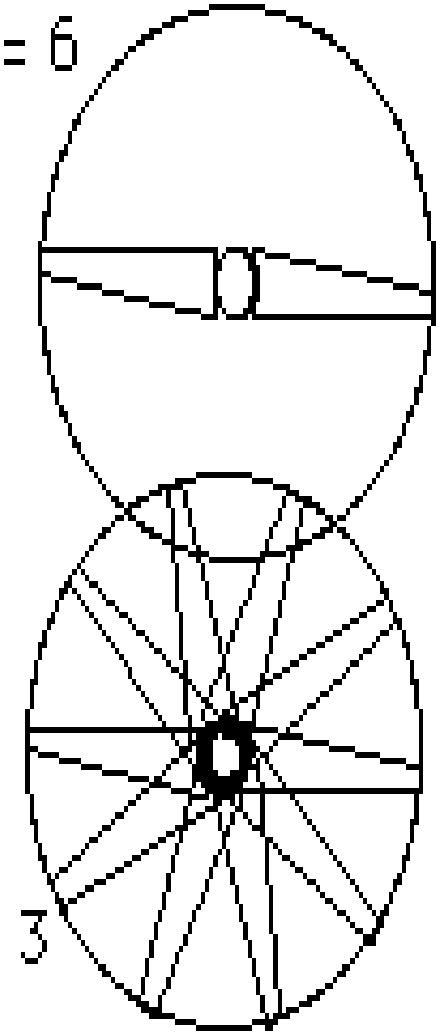
Blades Optimum TSR

No of Blades	TSR(λ)
• 1	Around 9
• 2	Around 6
• 3	Around 4—5
• 4	Around 3
• 6	Around 2
• 10	Around 3

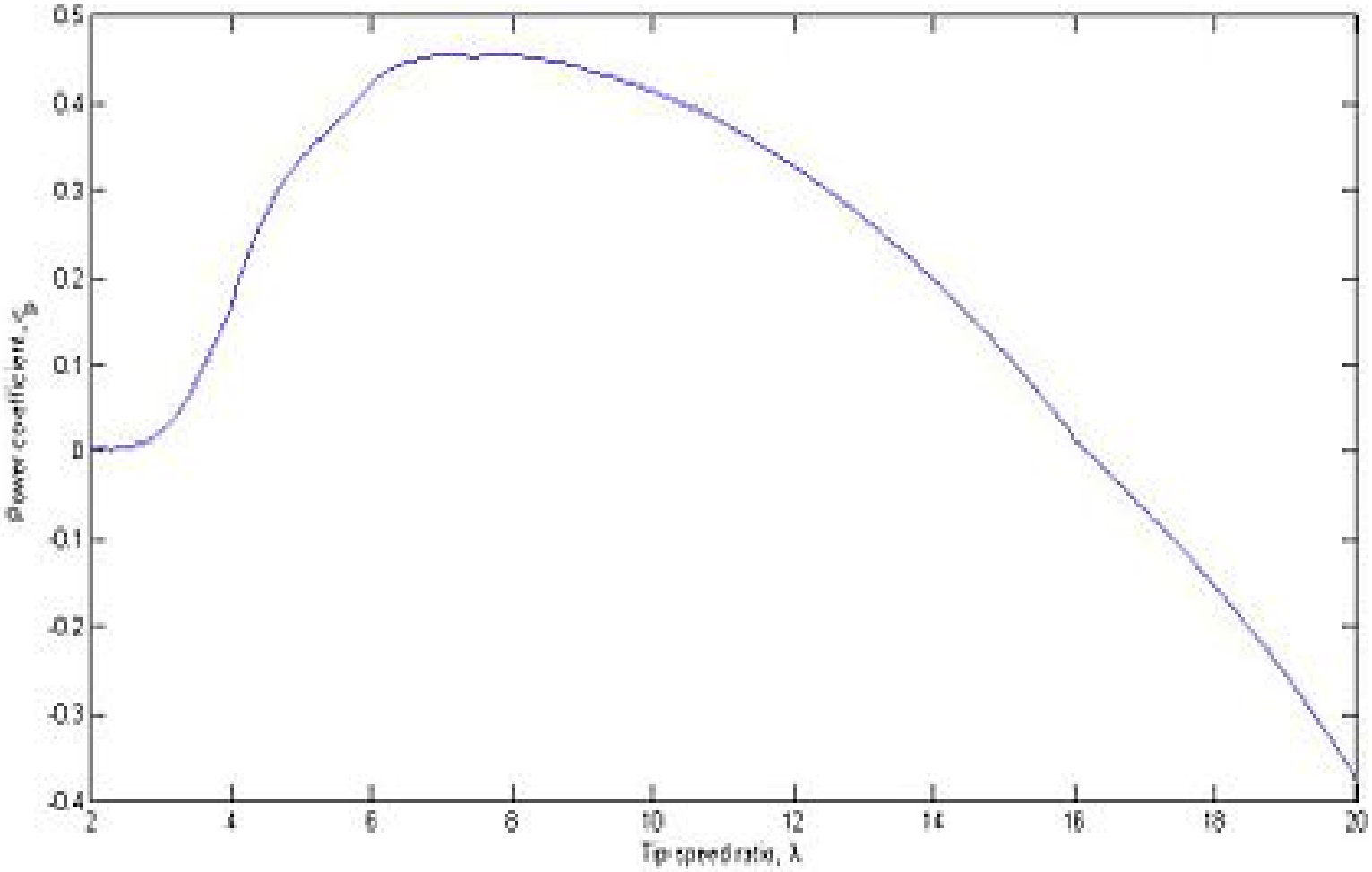
tip speed ratio λ:-



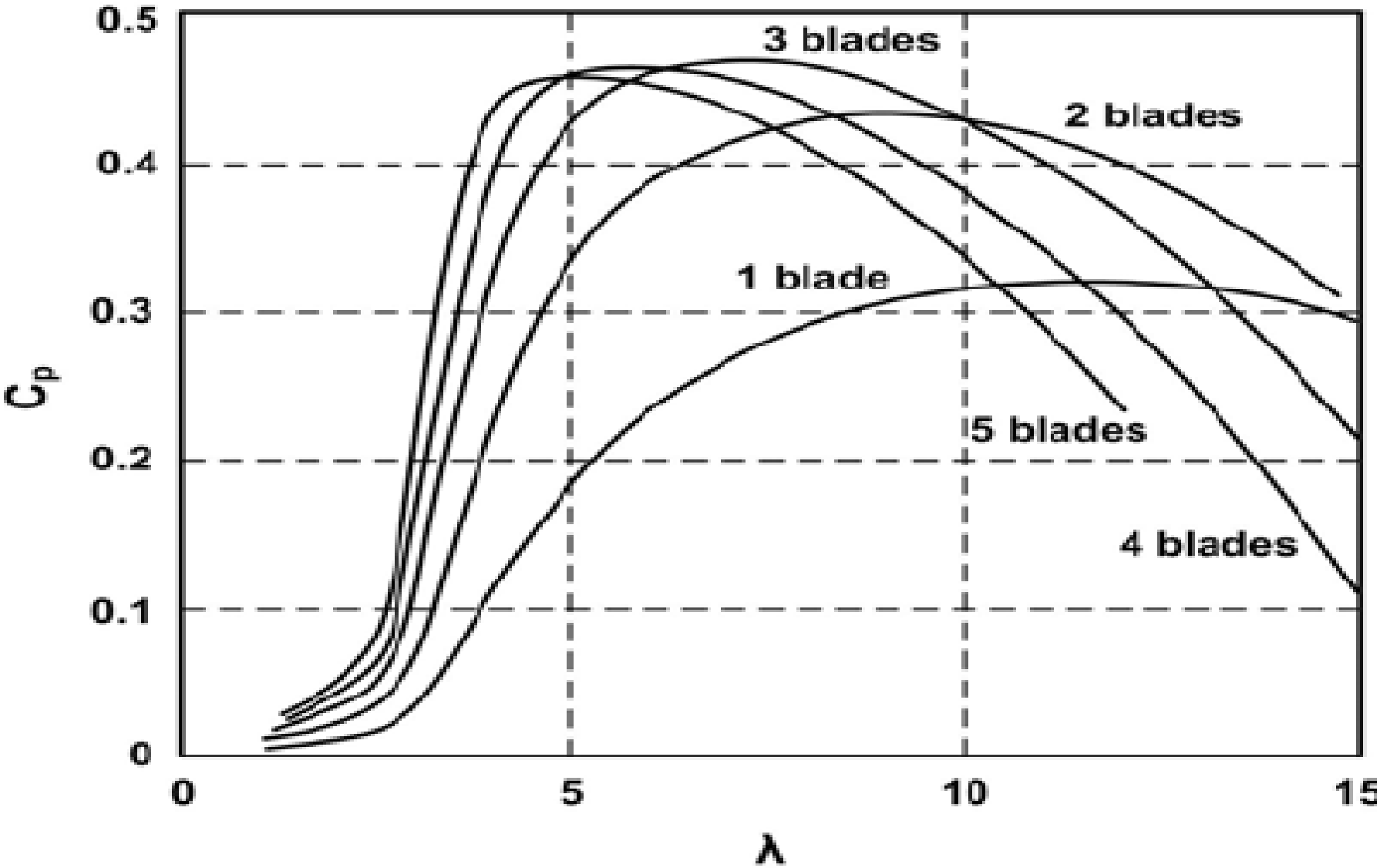
THE BLADE ANGLES ARE DIFFERENT IN EACH CASE. ONLY THE PLANFORM IS THE SAME.



Power Coefficient vs Tip Speed ratio



Cp vs λ


0.5
0.4
0.3
0.2
0.1
0
C_p
0
5
10
15
λ
3 blades
2 blades
1 blade
5 blades
4 blades

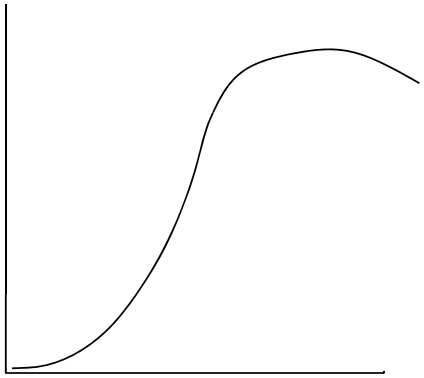
EFFECT OF ROTOR TIP SPEED RATIO

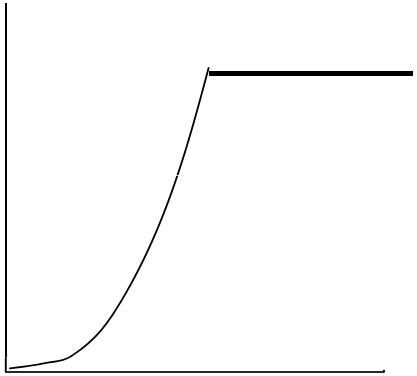
1. Rotor blade tips rotating at a speed larger than 80 m/sec will be subject to erosion of the leading edges from their impact with dust or sand particles in the air, and will require the use of special erosion resistant coatings much like in the design of helicopter blades.
2. Noise generation in the audible and non audible ranges.
3. Vibration, particularly in the cases of two or single bladed rotors.
4. Starting difficulties if the shaft is stiff to start rotation.
5. Reduced rotor efficiency due to drag and tip losses.
6. Excessive rotor speeds would lead to a runaway turbine, leading to its catastrophic failure, and even disintegration.

- The conclusion here is Tip speed ratio strongly affects the power.
- **Increasing the Tip speed ratio will increase the power**.
- As we move towards the tip of the blade of the wind turbine the wind speed is more.

Control Systems & Electronics

- Control methods
  - Drivetrain Speed
    - Fixed (direct grid connection) and Variable (power electronics for indirect grid connection)
  - Blade Regulation
    - Stall – blade position fixed, angle of attack increases with wind speed until stall occurs behind blade
    - Pitch – blade position changes with wind speed to actively control low-speed shaft for a more clean power curve

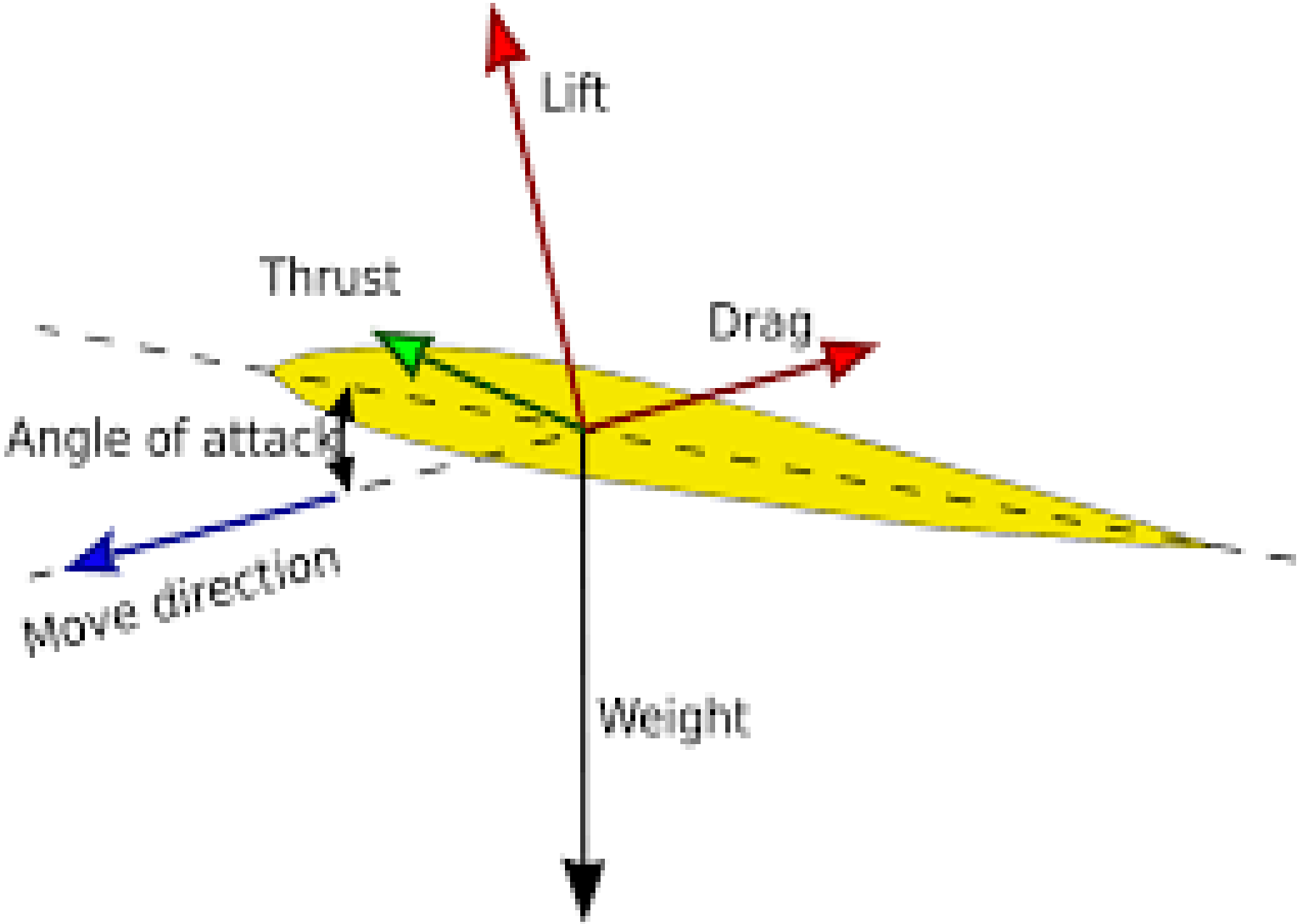




Control Unit

- The main purpose is to limit the power extracted at high wind speeds (above-rated speed) and to start the turbine below that speed.
- High speeds typically happen only for a short time, however, if they are not controlled they can/will damage the drive train, gearbox, and generator.
- Aerodynamic control is the most effective form.
  - Stall control (passive)
  - Pitch control (active)
  - Yaw control (active)

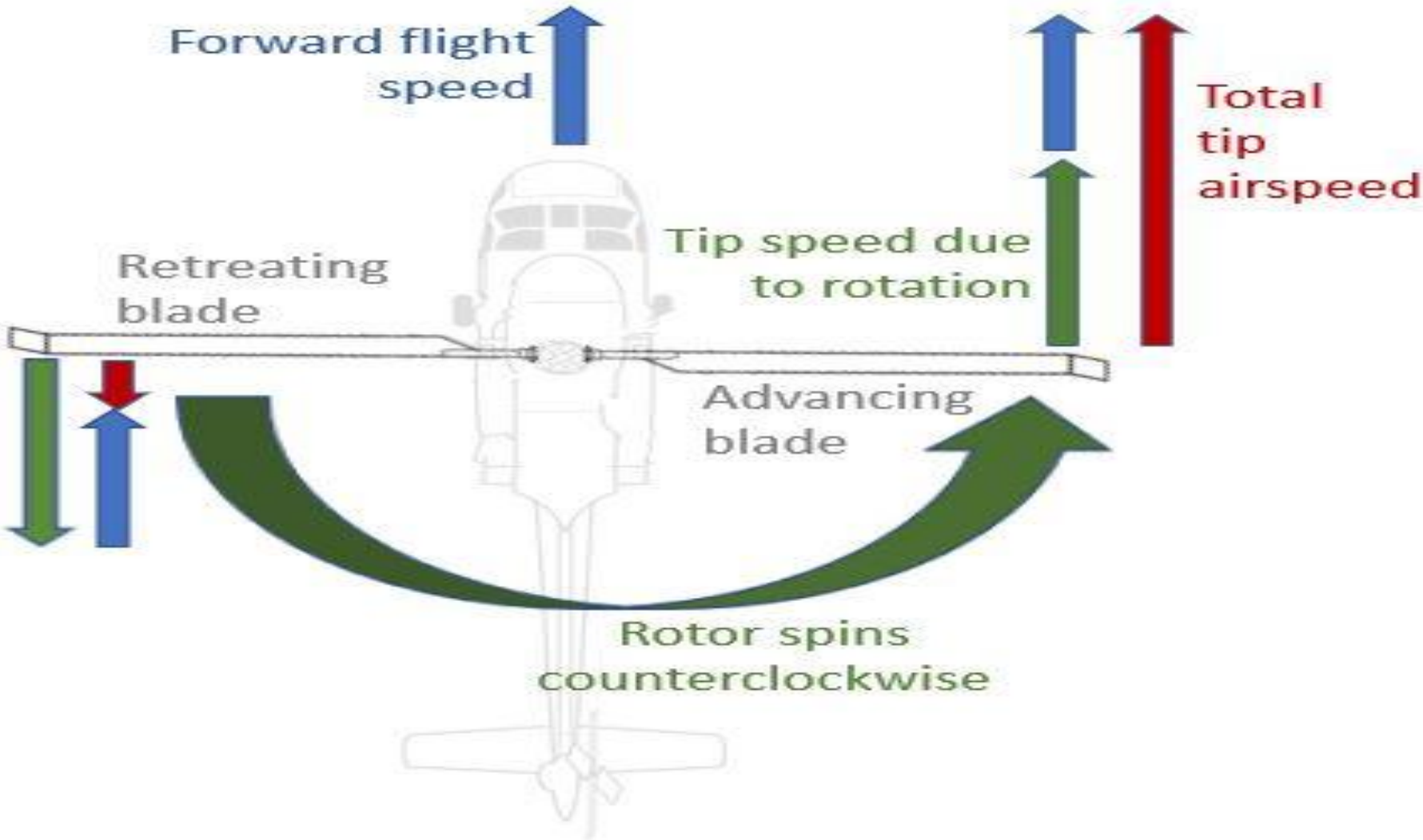
- **Lift force** – defined to be perpendicular to direction of the oncoming air flow.
- The lift force is a consequence of the unequal pressure on the upper and lower airfoil surfaces.
- **Drag force** – defined to be parallel to the direction of the oncoming air flow.
- The drag force is due both to viscous friction forces at the surface of the airfoil and to unequal pressure on the airfoil surfaces facing toward and away from the oncoming flow.


Lift
Thrust
Drag
Angle of attack
Move direction
Weight

Stall Control

- stall control is another approach for controlling and protecting wind turbines.
- The concept of stall control is that the power is regulated through stalling the blades after rated speed is achieved.
- Stall control can be further divided into passive and active control approaches.

Stalling a blade



Passive and Active Stall control

Passive

- The geometry of the rotor blade profile is aerodynamically designed to ensure that in high wind speed conditions, it creates turbulence on the side of the rotor blade which not facing the wind.
- This stall prevents the lifting force of the rotor blade from acting on the rotor.

Active

- active control provides more accurate control on the power output and maintains the rated power at high wind speeds.
- An active stall system reads the power output the way a pitch-controlled system does, but instead of pitching the blades out of alignment with the wind, it pitches them to produce stall.

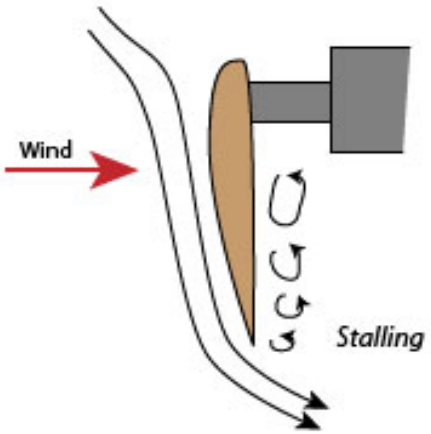
Control Mechansim

A control mechanism is an overspeed control that allows the rotors to be slowed down or stopped. Its purpose is to

- optimize aerodynamic efficiency
- keep the generator with its speed and torque limits and rotor and tower within strength limits
- enable maintenance
- educe noise

Stalling

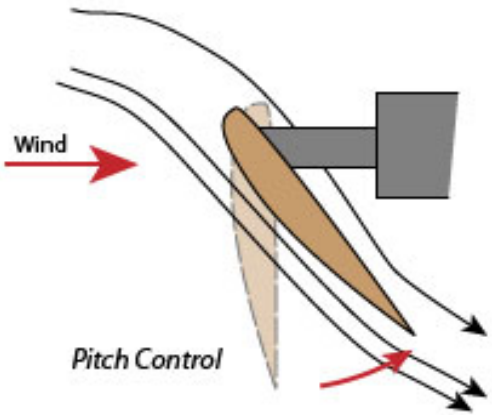
Principle: Increased angle of attack results in decreasing lift-to-drag ratio.



- Passive: Blades are at a fixed pitch that starts to stall when the wind speed is too high.
- Active: motor turns the blades towards stall when wind speeds are too high.
- Hybrid: Pitch can be adjusted manually to reflect site's particular wind regime.
- Disadvantage: Stalled blades cause large vibration and therefore noise.

Pitch Control

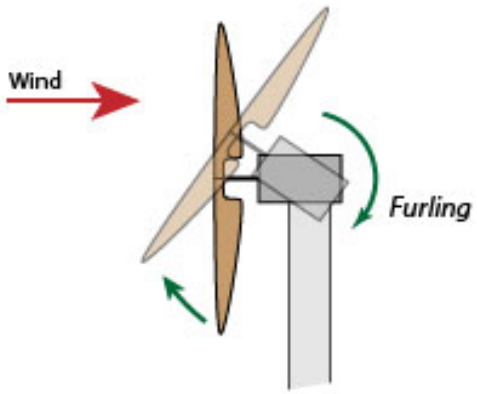
Principle: Decrease angle of attack also results in decreasing lif-to-drag ratio.



Always active control: Blades rotate out of the wind when wind speeds are too high.

Furling

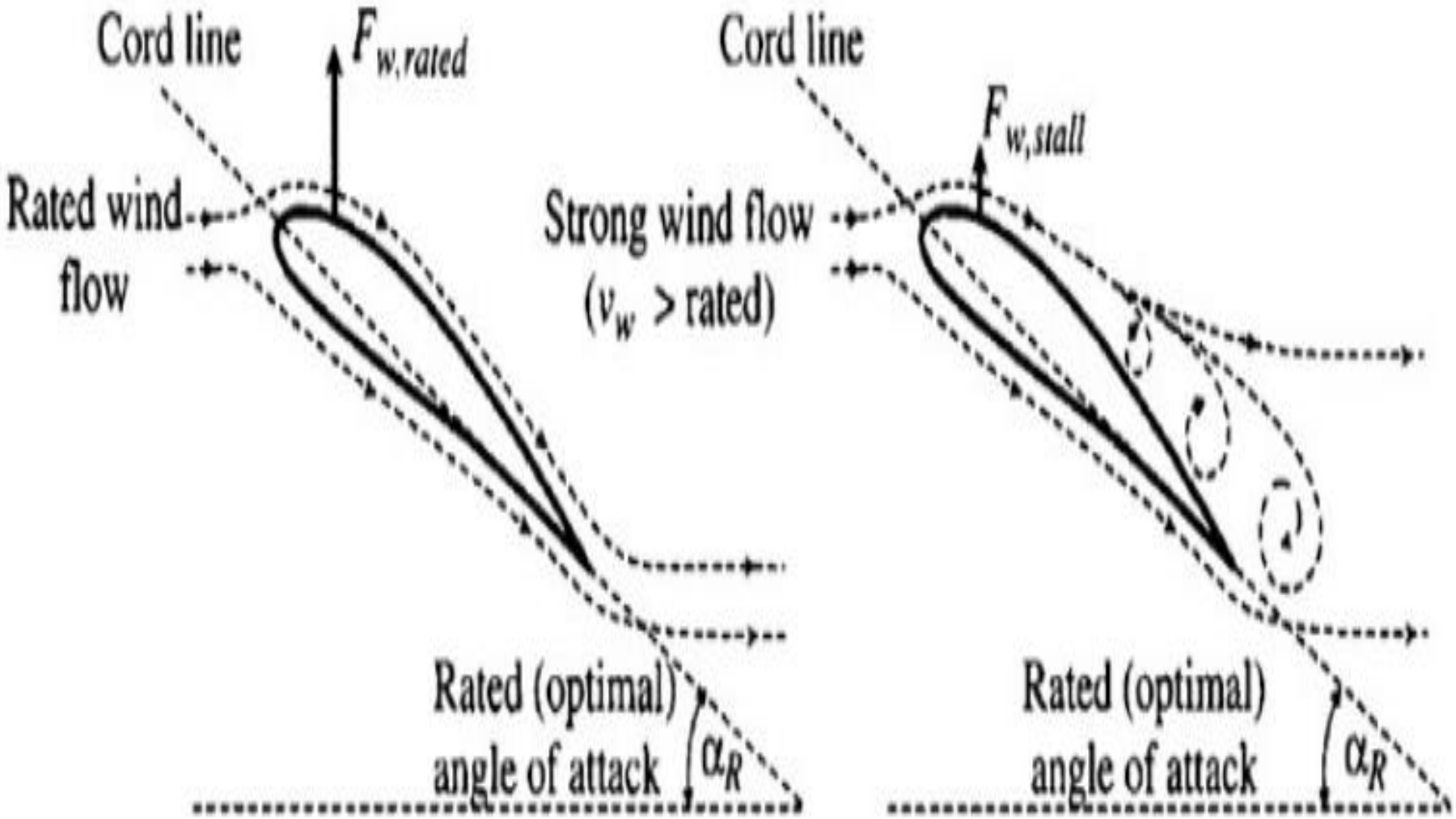
Principle: Moving the axis out of the direction of the wind decreases angle of attack and cross-section



Standard modern turbines all furl in high wind.

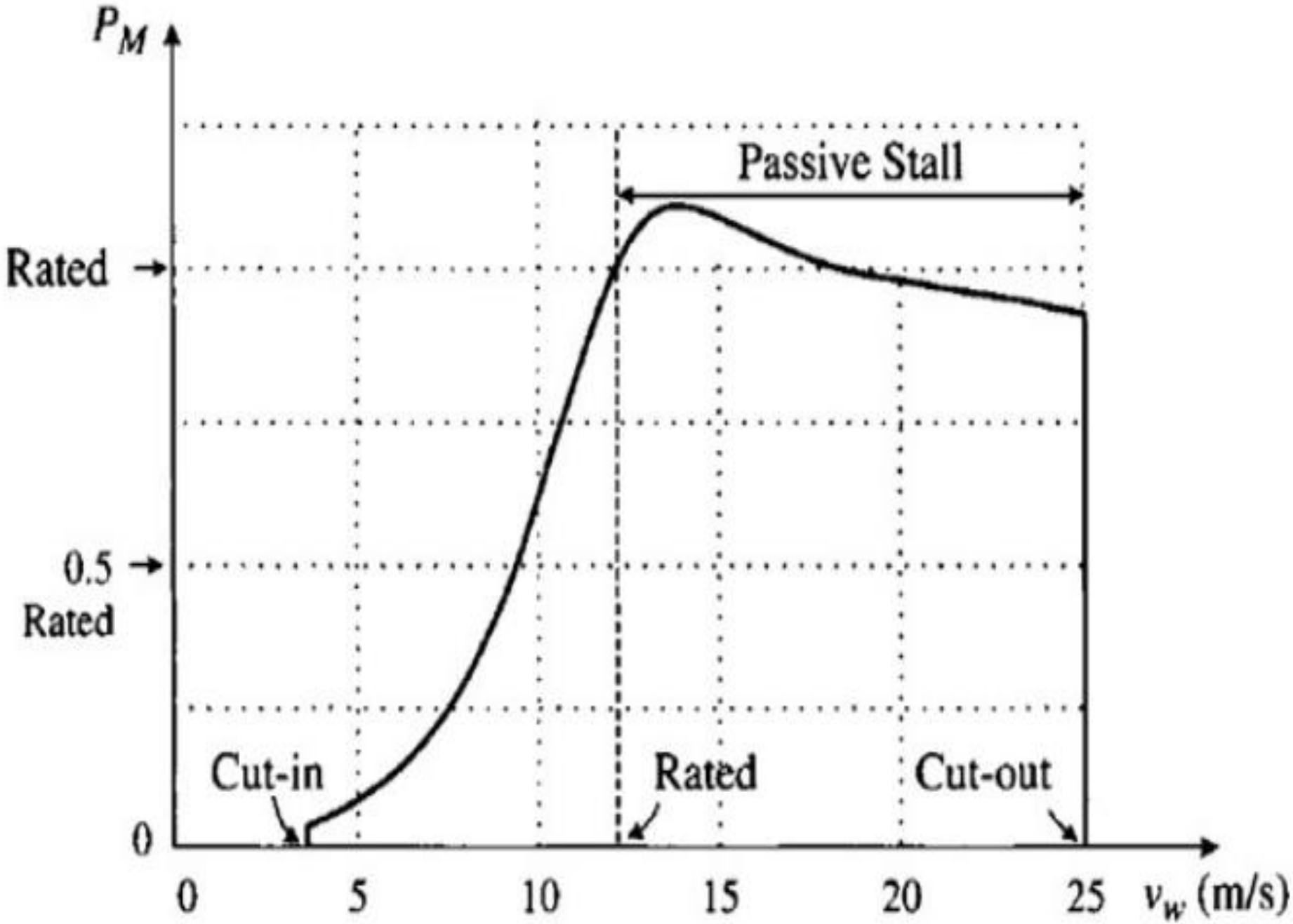
- Requires active pitch control: Pitch angle of the blades needs to be minimised first, otherwise the torque on the rotor would be to big for furling.
- Active: Vertical furling (as diagram) with hyrdraulic, spring-loaded or electric motor driven.
- Passive: Horizontal furling with yaw

Activate Wi
Go to PC setting



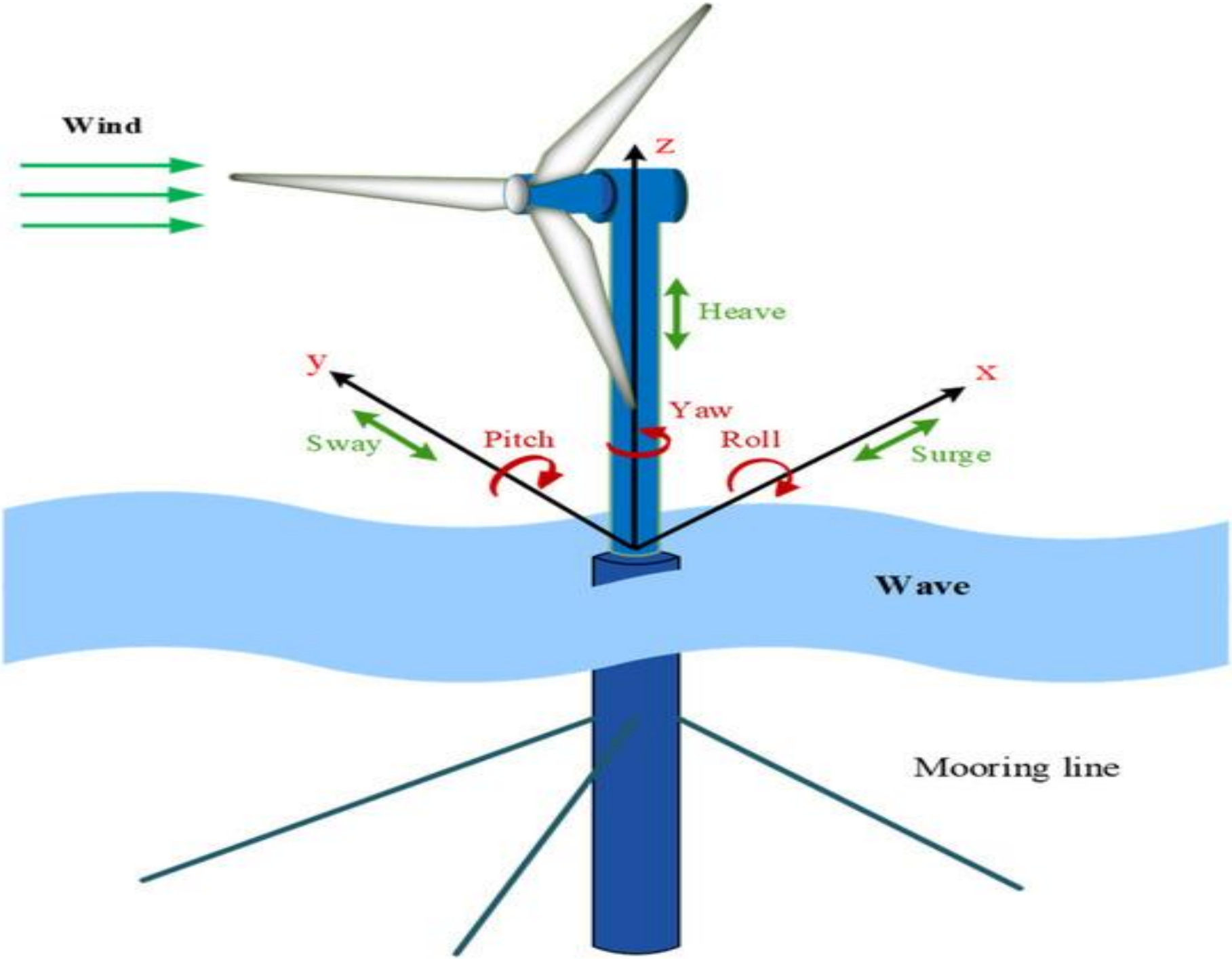
(a) At rated wind speed

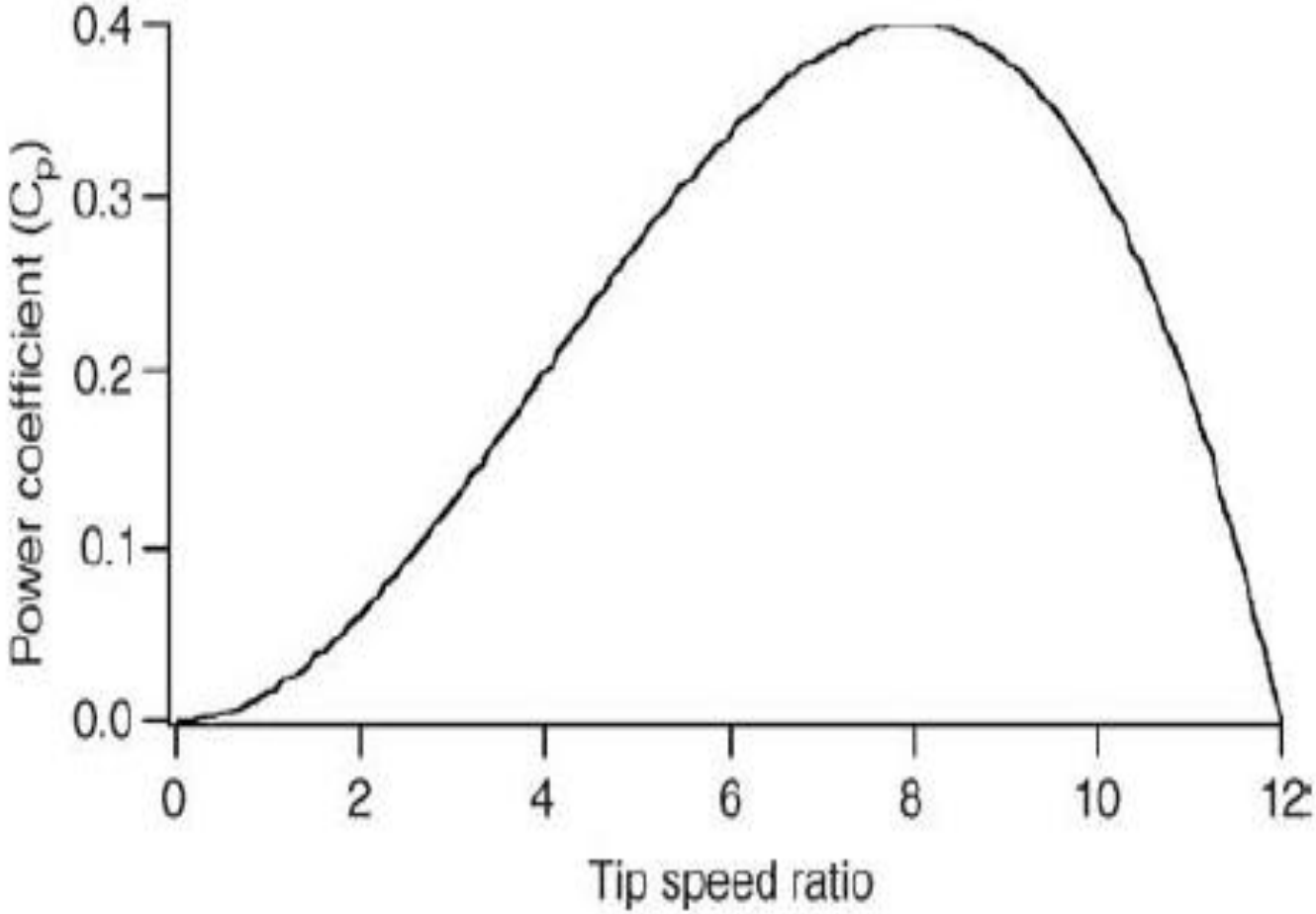
(b) Above rated wind speed


P_M
Passive Stall
Rated
0.5
Rated
Cut-in
Rated
Cut-out
0
0
5
10
15
20
25
v_w (m/s)

Pitch control

- A pitch control system's goal is to both monitor and adjust a wind turbine's rotor blade angle.
- These blades which can be as long as 65 meters or more and are responsible for the turbine's rotational speed.


Wind
z
Heave
y
x
Yaw
Sway
Pitch
Roll
Surge
Wave
Mooring line


0.4
0.3
0.2
0.1
0.0
Power coefficient (C_p)
0
2
4
6
8
10
12
Tip speed ratio

Pitch angle control

- Blade pitch used for constant rotor speed.
- Use to help control average power.

- Change orientation of the blades to change the aerodynamic forces.

- Blades are pitched towards incoming wind as wind speed increases.

  Angle of attack decreases

  Forces decrease

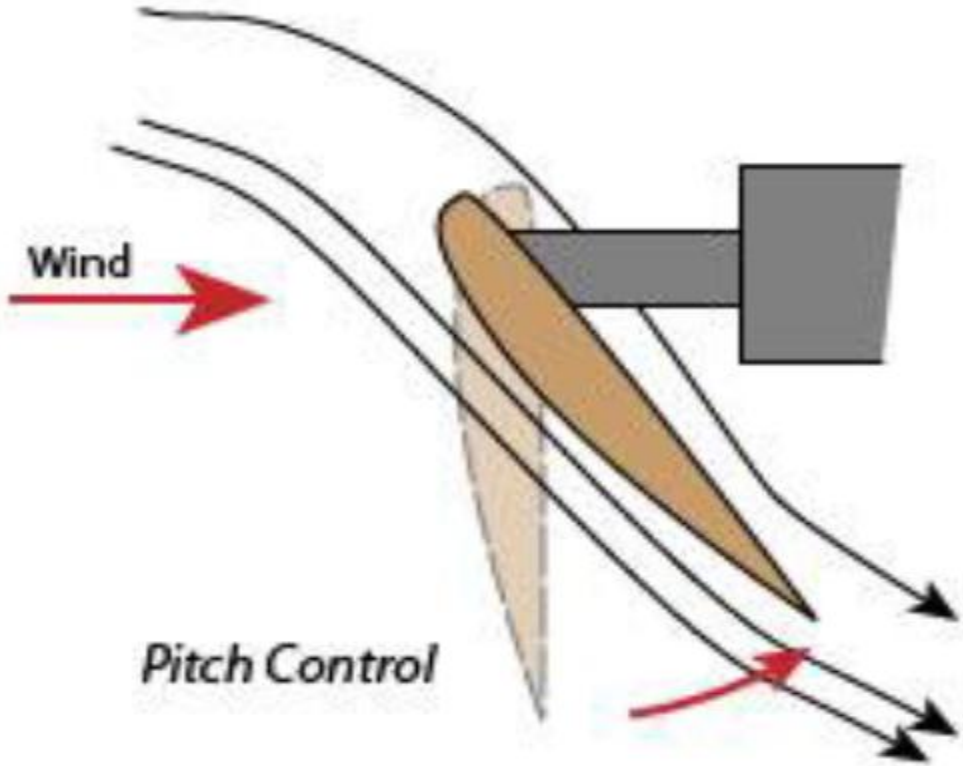
  Sheds power

- We can use pitch adjustment to stall and furl, two methods of pitch control.
- By stalling a wind turbine, **you increase the angle of attack**, which causes the flat side of the blade to face further into the wind.
- Furling **decreases the angle of attack**, causing the edge of the blade to face the oncoming wind.

- Stall control
  - Wind turbine stops if wind speed is out of the range.
- Active stall control
  - Wind turbine adjusts to stall if wind speed is out of the range.
- Pitch angle control
  - Wind turbine adjusts to stall and furl if the wind speed is out of the range.

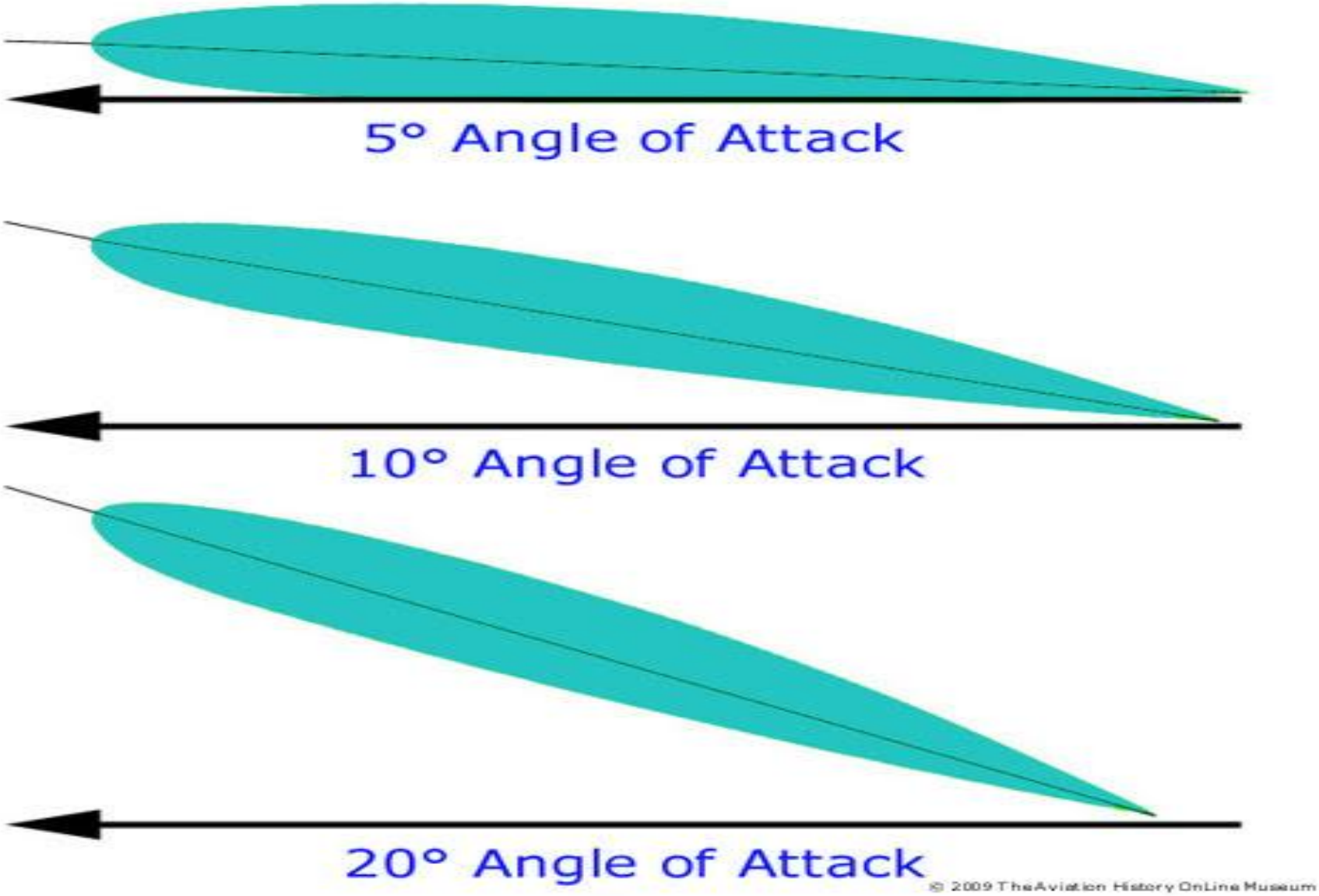
Pitch angle control

***Pitch Control Principle*:** Decrease angle of attack also results in decreasing lift-to-drag ratio.

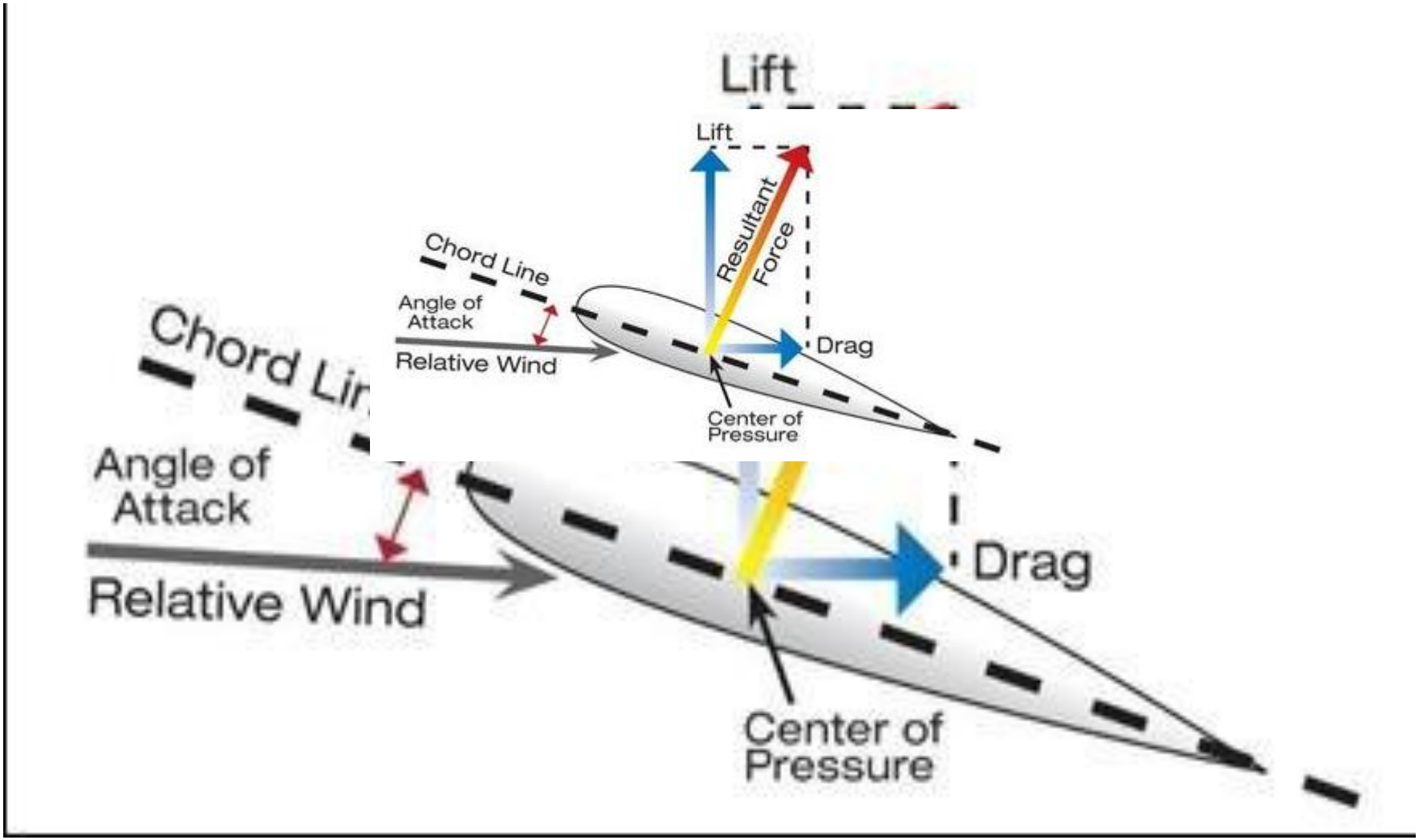


The schematic representing the pitch control regulator.

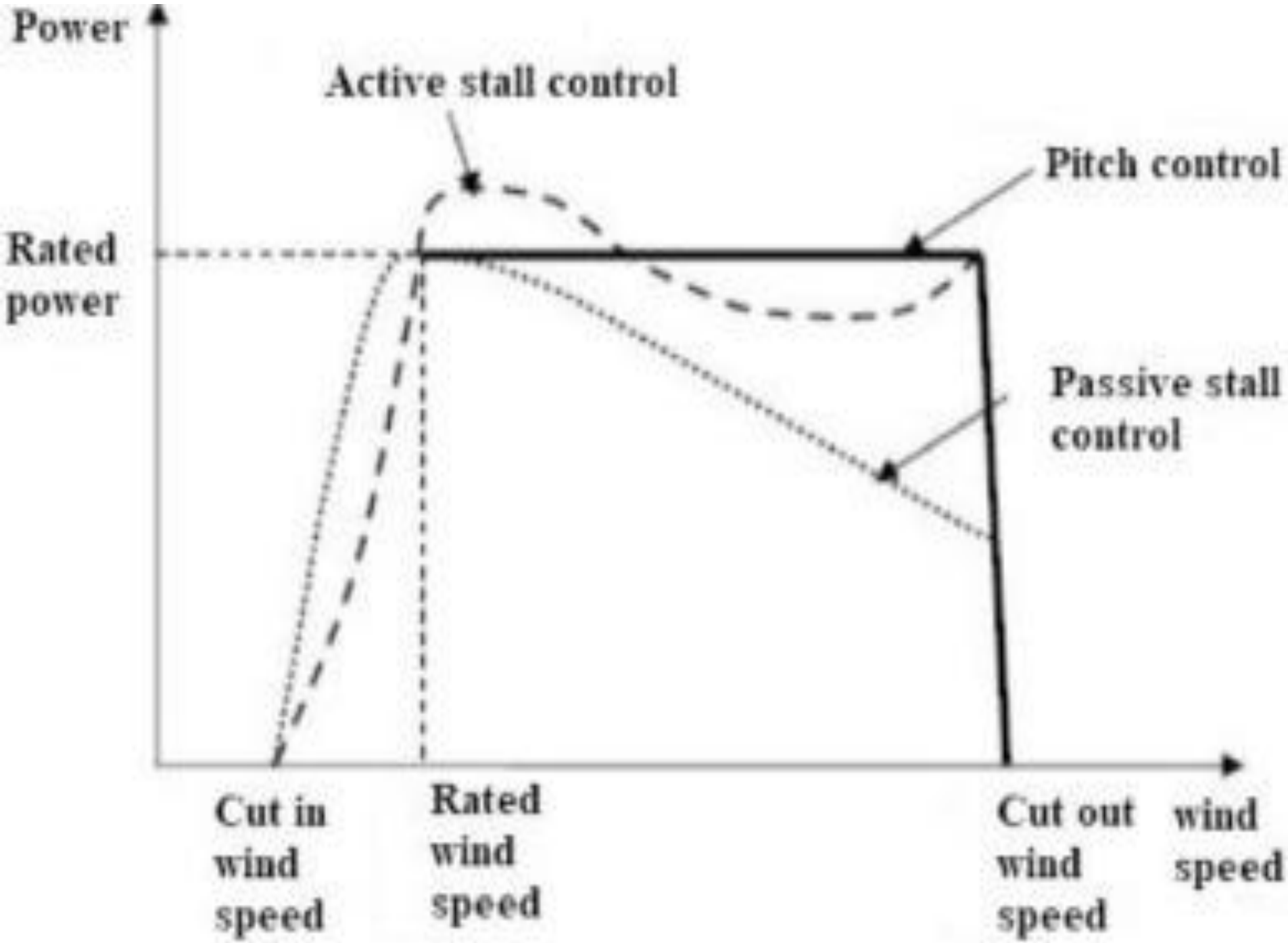
Angle of Attack



Lift and Drag force of wind



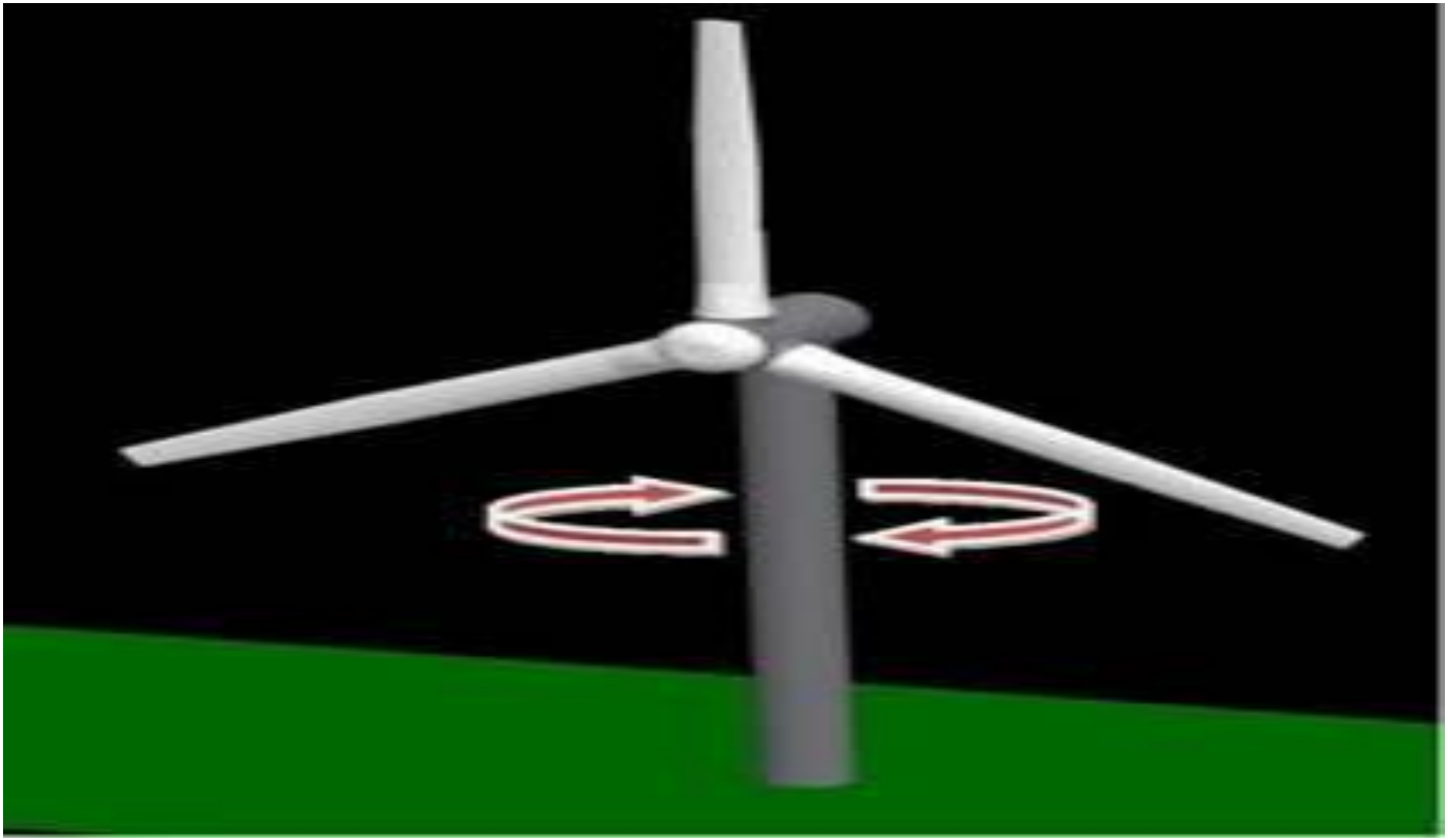
Blade 3
Variable propeller
Wind-speed
measuring instrument
Blade 1
Directions
ALSTOM
Wind turbine hub
Wind turbine nacelle
Blade 2
Tower


Power
Active stall control
Pitch control
Rated power
Passive stall control
Cut in wind speed
Rated wind speed
Cut out wind speed
wind speed

Pitch Drive System

- Electric
- hydraulic

Yaw control



- Passive yaw
- Active yaw
  - Yaw drive
    - Yaw motor
    - Yaw bearing
    - Yaw gear
    - Yaw motor with gear box

Grid Integration Issues

- Output Power Prediction
- Reactive Power/Voltage Support
- Impact of the Frequency
- Impact of the Harmonics/Power Quality Issues
- Angular Stability/Inter-Area Oscillation
- Grid Reliability and Resiliency
- Protection Challenges
- Low Voltage Ride-Through (LVRT) Capability
- Planning Challenges
- Transmission, Communication, and Security Challenges
- Electricity Market Challenges
- Socio-Economic and Environmental Challenges

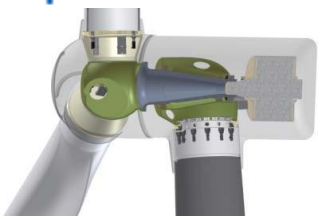
Solutions Adopted to Mitigate Integration Effects

- Grid Codes
- Advanced Monitoring and Control Strategies
- Energy Storage Systems
- Renewable Energy Policies

Future Technology Development

- Improving Performance:
  - Capacity: higher heights, larger blades, superconducting magnets
  - Capacity Factor: higher heights, advanced control methods (individual pitch, smart-blades), site-specific designs
- Reducing Costs:
  - Weight reduction: 2-blade designs, advanced materials, direct drive systems
  - Offshore wind: foundations, construction and maintenance

American Superconductor®

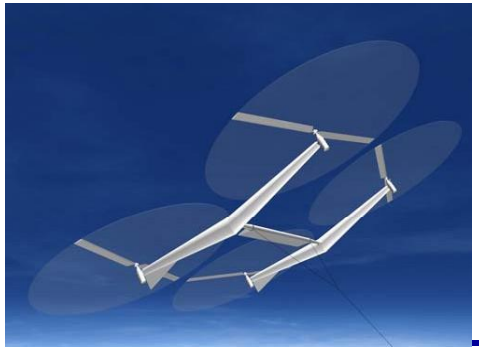


VERGNET GROUPE

Northern POWER SYSTEMS

Future Technology Development

- Improving Reliability and Availability:
  - Forecasting tools (technology and models)
  - Dealing with system loads
    - Advanced control methods, materials, preemptive diagnostics and maintenance
  - Direct drive – complete removal of gearbox
- Novel designs:
  - Shrouded, floating, direct drive, and high-altitude concepts

FloDesign Wind Turbine
there's change in the wind

Sky Windpower